

PHARMACEUTICAL ANALYSIS II

Chapter 1 **UV- Visible spectrophotometry**

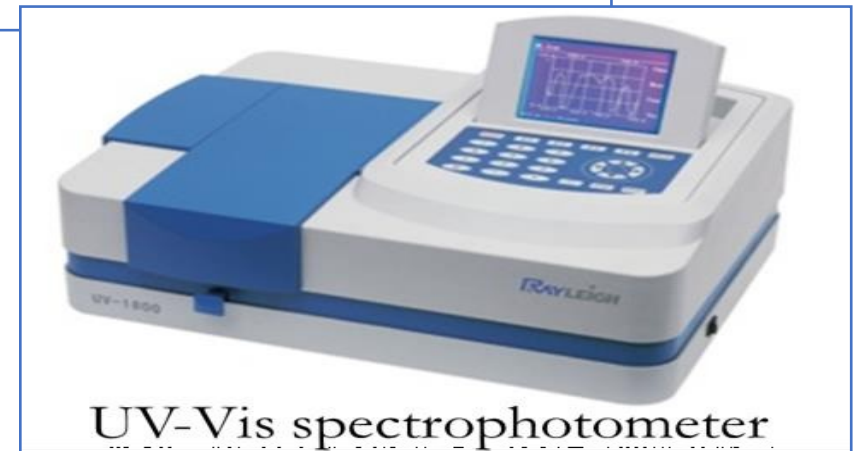
❖ At the end of this lesson, student's will be able to:

- Define UV-Visible spectroscopy.
- Describe the parameters for quantification of EM radiation as a wave
- Explain the particle properties of Light
- Discuss the Interaction between EMR with Matter
- Explain the principles of UV-Vis Spectrophotometry
- Describe the instrumentation of UV-Vis spectrophotometry
- Discuss the applications of UV-Vis Spectrometry in pharmaceutical analysis
- Explain the factors governing absorption of radiation in the UV/Visible region
- Discuss the advantages and disadvantages of UV-Vis Absorption Spectroscopy

OUT LINE

- **UV-Visible Spectrophotometry**

- Introduction
- Factors governing absorption of radiation in the UV/Visible region
- Instrumentation
- Qualitative spectrophotometry
- Quantitative spectrophotometry
- Applications in Pharmaceutical Analysis
- Colorimetry



UV-Vis spectrophotometer

Introduction to Spectroscopy

Spectroscopy

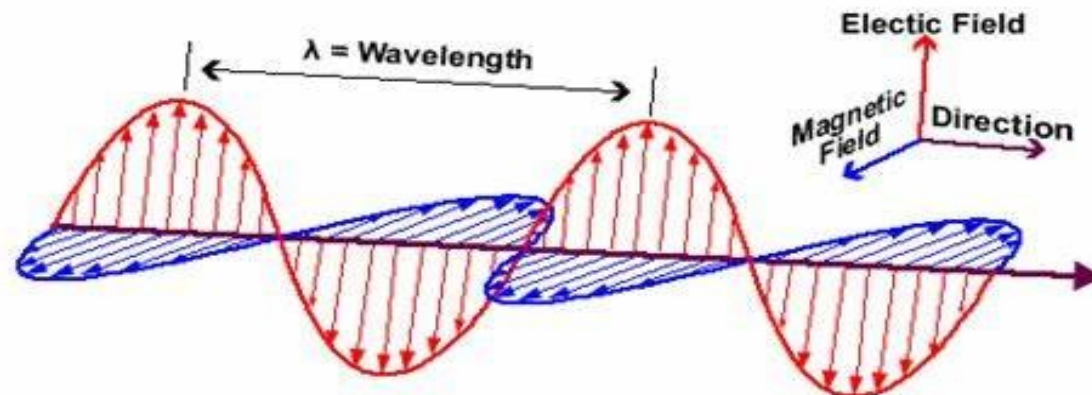
- Spectroscopy is one of the valuable technique used to study **atomic & molecular structure**
- **Spectroscopy:** is the study of interaction of **electromagnetic radiations (EMR)** with atoms or molecules (with matter)
- The consequence of this interaction is energy gets **absorbed or emitted** by atoms or molecules
- It is of **two types**:
 - **Emission spectroscopy**
 - **Absorption spectroscopy**

Introduction to Spectroscopy

- **Emission spectroscopy**
 - It is the measurement of emitted light
 - Example: Flame photometry, Fluorimetry
- **Absorption spectroscopy**
 - It is the measurement of energy absorbed by an excited atom or molecule
 - Example: UV-Visible, IR, NMR
- **Spectroscopy can be also be classified as:**
 - **Atomic Spectroscopy:**
 - It deals with the interaction of EMR with atoms
 - **Molecular Spectroscopy:**
 - It deals with the interaction of EMR with molecules

Introduction to Spectroscopy

- Spectroscopy is the study of interaction between electromagnetic radiation and **matter**.
- **EMR** is a form of energy whose behavior is described by the properties of both **waves** and **particles**.
- EMR consists of oscillating electric and magnetic fields that propagate through space along a linear path and with a constant velocity.



Introduction...

❖ Electromagnetic waves are usually described in terms of:

(a) wavelength (λ), the distance between two successive peaks or trough.

Different units of length are used to express wavelengths

E.g. Angstrom, centimeter, micro and nanometer

$$\checkmark 1 \text{ m} = 10^2 \text{ cm} = 10^3 \text{ mm} = 10^6 \mu\text{m} = 10^9 \text{ nm} = 10^{10} \text{ \AA}.$$

$$\checkmark 1 \text{ \AA} = 10^{-1} \text{ nm} = 10^{-4} \mu = 10^{-7} \text{ mm} = 10^{-8} \text{ cm} = 10^{-10} \text{ m}$$

Introduction...

- ✓ For ultraviolet and visible electromagnetic radiation the wavelength is usually expressed in **nanometers (nm, 10^{-9}m)**,
- ✓ the wavelength for infrared radiation is given in **microns (μm , 10^{-6}m)**.

(b) wave number (n , ν'), the number of waves per centimeter.

$$\checkmark \bar{\nu} = 1/\lambda$$

(c) frequency (ν), the number of waves per second.

(d) Amplitude (A), is the vertical distance from midline of a wave to the peak or trough.

Measured by units of distance

(e) Velocity is a product of wavelength and frequency

Introduction...

- ❖ The wave model fails to account for phenomenon associated with the absorption and emission of radiant energy.
- EMR can be treated as discrete packets of energy or particles called **photons** or **quanta**.
- ✓ the energy of a photon is directly proportional to its frequency.
- ✓ $E = h\nu = hc/\lambda = hc\bar{\nu}$, Where h is Planck's constant, which has a value of $6.626 \times 10^{-34} \text{ J} \cdot \text{s}$.
- ✓ velocity, c - the speed of propagation, $c = 2.9979 \times 10^8 \text{ m/s}$ (in vacuum)

Example

- 1) Calculate the wavenumber of a beam of infrared radiation with a wavelength of $5.00 \mu\text{m}$

– Solution:

$$\bar{\nu} = \frac{1}{\lambda} = \frac{1}{5.00 \cancel{\mu\text{m}} \times 10^{-4} \text{ cm}/\cancel{\mu\text{m}}} = 2000 \text{ cm}^{-1}$$

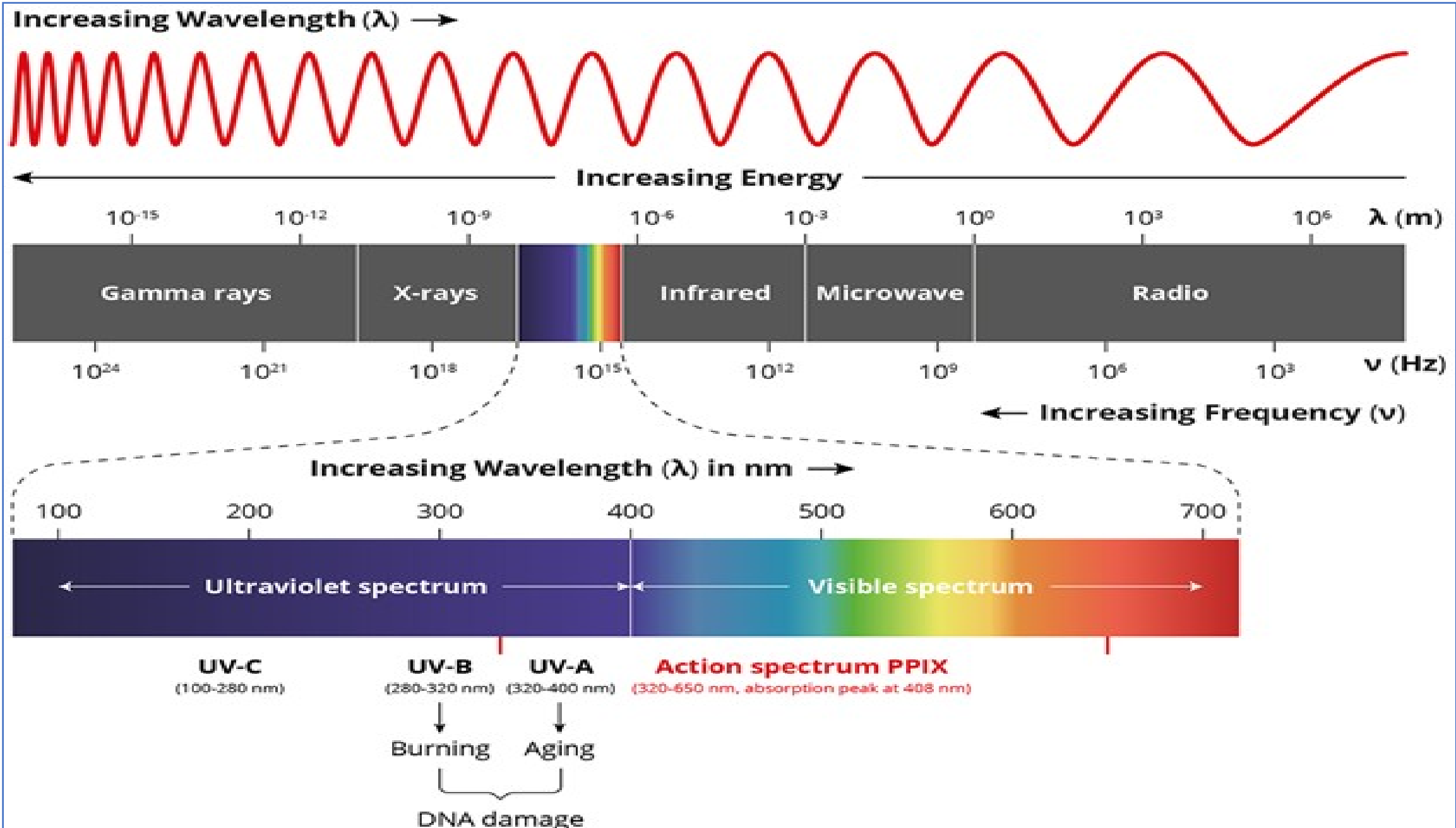
- 2) Calculate the energy in joules of one photon of radiation with the wavelength given in Example -1. (Use $C = 3 \times 10^{10} \text{ cm/s}$)

– Solution:

$$E = hc\bar{\nu} = 6.63 \times 10^{-34} \text{ J} \cdot \cancel{s} \times 3.00 \times 10^{10} \frac{\text{cm}}{\cancel{s}} \times 2000 \text{ cm}^{-1}$$
$$= 3.98 \times 10^{-20} \text{ J}$$

Electromagnetic Spectrum

- The electromagnetic spectrum covers an enormous range of energies (frequencies) and thus wavelengths.
- ✓ based on the type of atomic or molecular transition that gives rise to the absorption or emission of photons.
- The boundaries are not rigid, and an overlap between spectral regions is possible.



How does EMR interact with Matter?

- Absorption

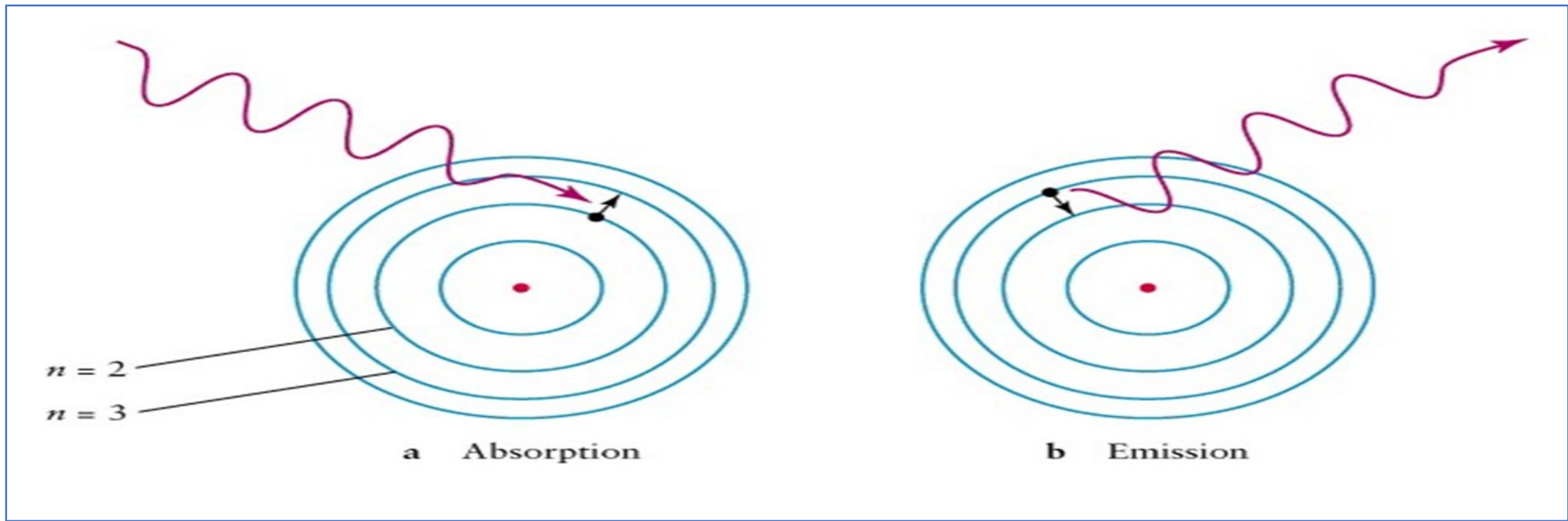
- ✓ EM energy transferred to absorbing molecule (transition from low energy to high energy state)

- Emission

- ✓ EM energy transferred from emitting molecule to space (transition from high energy to low energy state)

- Scattering

- ✓ Redirection of light with no energy transfer



- Absorption of UV/Visible affects all types of motions, the major effect being transition of valence electrons.
- Absorption of IR radiation results in Vibrational and rotational energy.

How does....?

- The excitation can occur at different degrees
 - Low energy tends to excite the outmost e^- 's first
 - An e^- at its excited state is not stable and tends to return its ground state.
 - If an e^- jumped more than one energy level because of absorption of a high energy, the process of the e^- returning to its ground state may take several steps, - i.e. to the nearest low energy level first then down to next.

How does...?

❑ An excited molecular species are extremely short lived and deactivation occurs due to:

1. Radiative Decay:

- excited molecules radiate energy as **photons**.
- When an electron transitions from a higher energy level to a lower one (from excited state to ground state), it emits a photon.
- The emitted light can be visible (**fluorescence**) or invisible (**phosphorescence**).

How does...?

2. Non-Radiative Processes:

- Not all energy release involves photons.
- ✓ **Vibrational Relaxation:** Molecules collide with their surroundings.
 - Vibrational relaxation is where the energy deposited by the photon into the electron is given away to other vibrational modes as kinetic energy.
 - This kinetic energy may stay within the same molecule, or it may be transferred to other molecules around the excited molecule.

How does...?

- This process is also very fast, between 10^{-14} and 10^{-11} seconds.
 - likely to occur immediately following absorbance.
- This relaxation occurs between vibrational levels,
- Electrons will not change from one electronic level to another through this method.
- ❖ if vibrational energy levels strongly overlap electronic energy levels, a possibility exists that the excited electron can transition from a vibration level in one electronic state to another vibration level in a lower electronic state

How does....?

- **Internal Conversion:** An electron hops between closely spaced energy levels within the same electronic state.
 - ✓ energy redistributes internally. the excess energy is redistributed among the molecule's vibrational and rotational modes.
 - ✓ The energy is “spread out” within the molecule, preventing it from remaining in the excited state.

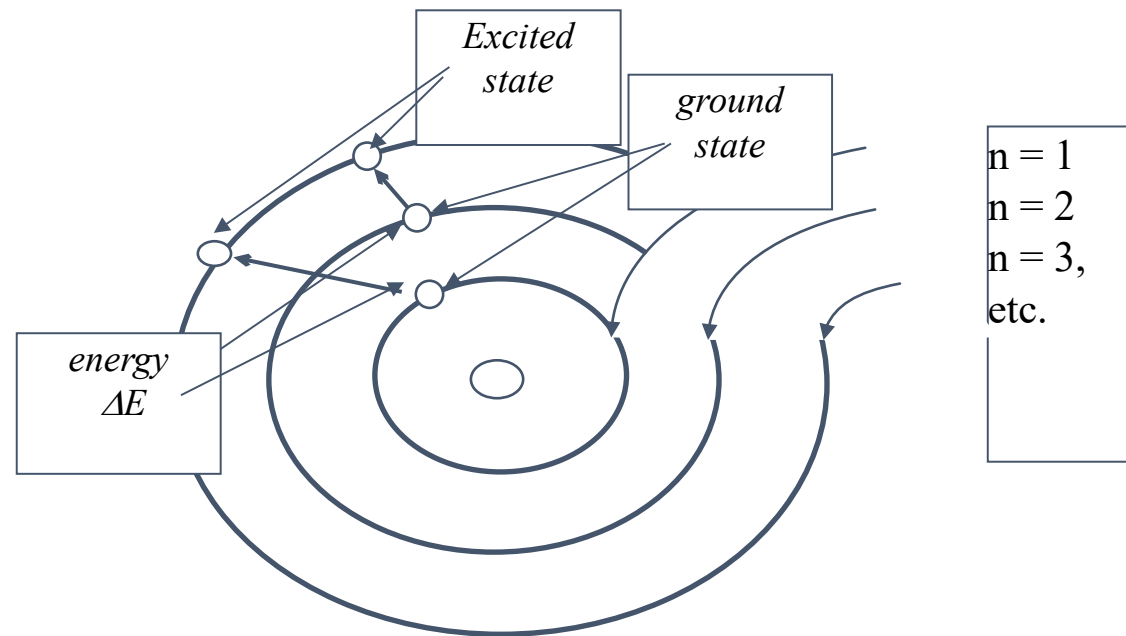
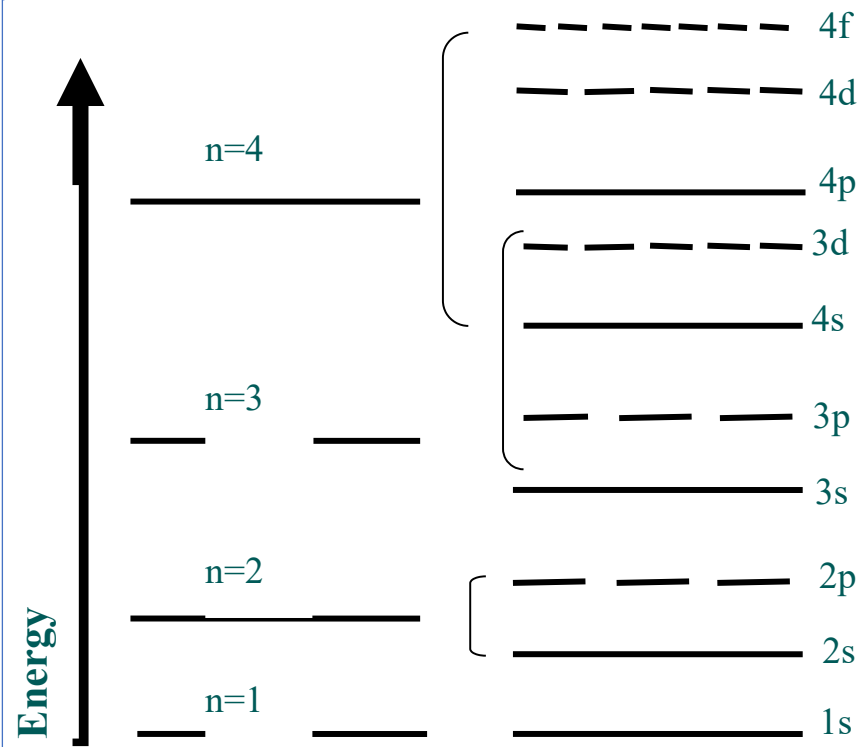
How does....?

3. Collisional Quenching:

- Excited molecules collide with other molecules (solvent, gas, etc.).
- These collisions can quench the excited state by transferring energy to the surroundings.
- For instance, a fluorescent dye in solution may lose its excited state energy to solvent molecules.

4. Chemical Reactions:

- Excited species can participate in chemical reactions.



How does...?

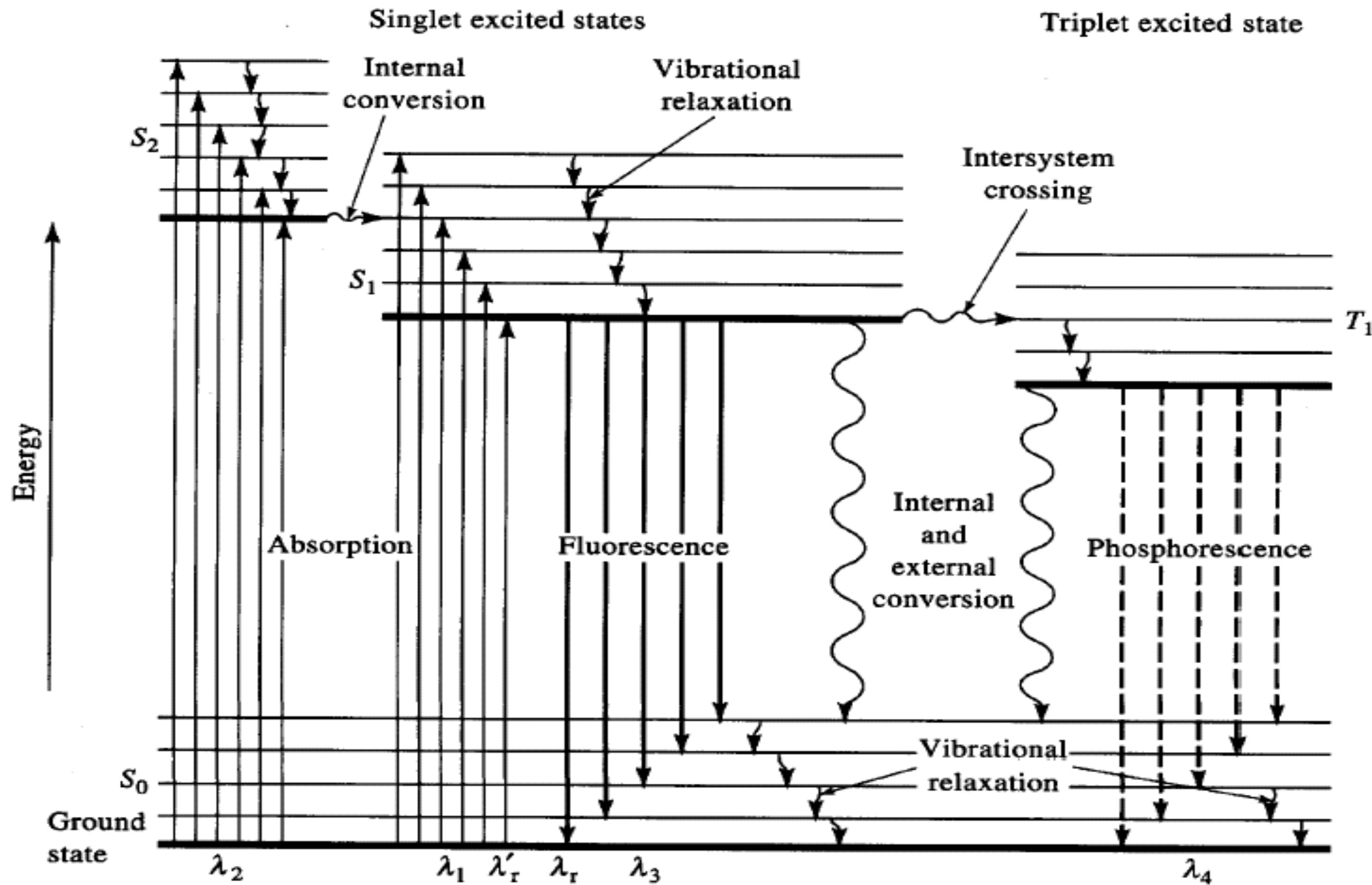
- ✓ Because light is a form of energy, absorption of light by matter causes the energy content of the molecules (or atoms) to increase.
- ✓ The total potential energy of a molecule generally is represented as the sum of its electronic, vibrational, rotational energies and other energies:
 - ❖ **$E = E_{\text{electronic}} + E_{\text{vibrational}} + E_{\text{rotational}}$**
 - $E_{\text{electronic}} > E_{\text{vibrational}} > E_{\text{rotational}}$**

Excitation of a Molecule

The energy levels of a molecule at each state / sub-state are *quantised*.

✓ To excite a molecule from its ground state (S_0) to a higher E state (S_1 , S_2 , etc.), *the exact amount of energy equal to the difference between the two states.*

$$h\nu' = E_{S_2, \nu_2} - E_{S_0, \nu_1}$$



UV/visible spectroscopy

- A spectroscopic technique which utilizes the UV/Visible region of the EMR is known as UV/visible spectroscopy/spectrophotometer/.
- ✓ Near UV region-200 nm - 400 nm
- ✓ Visible region- 400-800 nm
- Absorption of light in these regions mainly causes electronic transition.

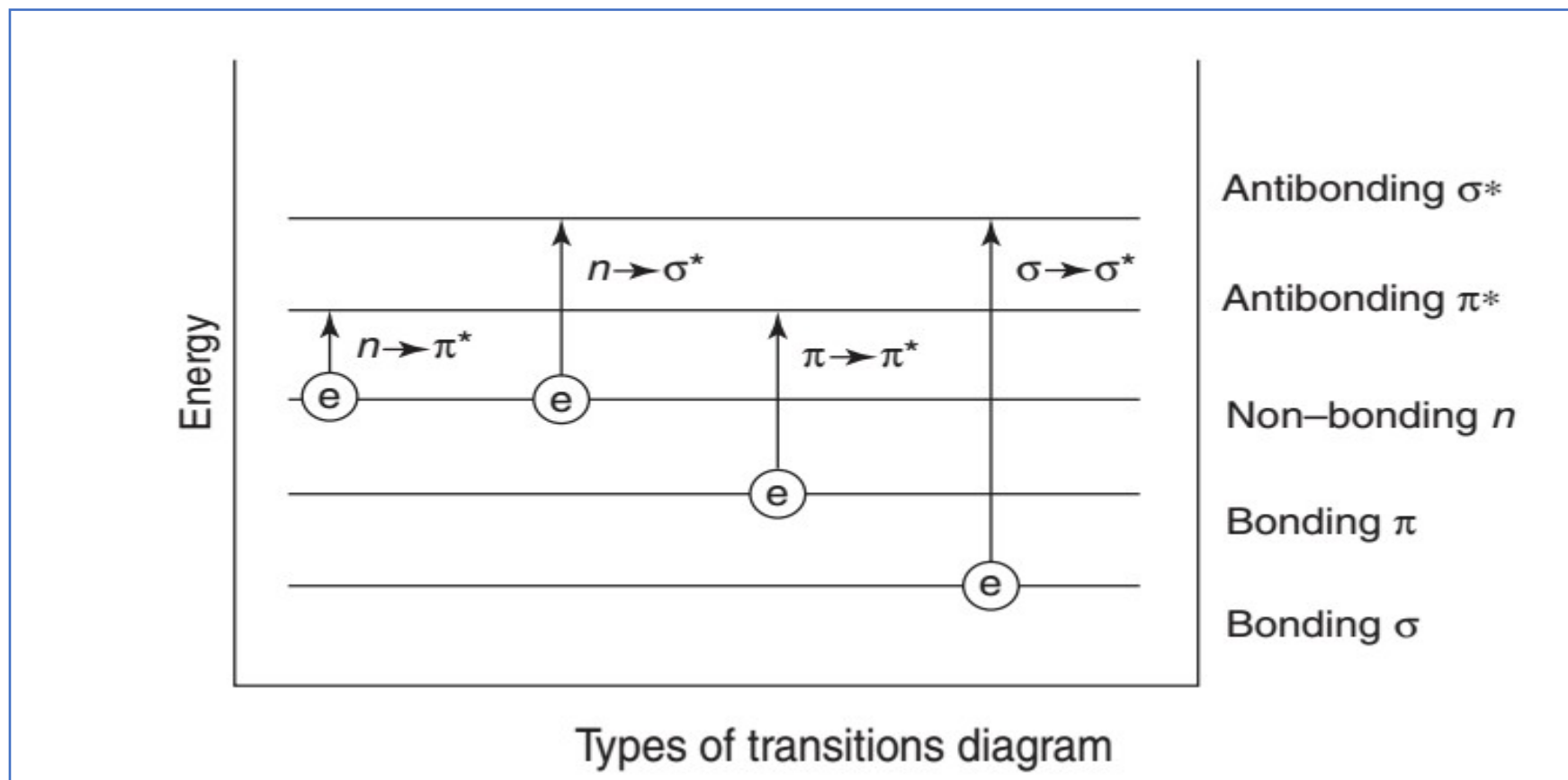
Electronic Transitions

❖ The outer electrons in an organic molecule may occupy one of the three different energy levels (σ - , π or n energy level).

□ Accordingly there are three types of electrons.

1. σ -electrons; They are bonding electrons, represent valence bonds and possess the lowest energy level (the most stable)
 2. π -electrons; They are bonding electrons, forming the pi-bonds (double bounds), and possess higher energy than sigma-electrons.
 3. n -electrons; They are nonbonding electrons, present in atomic orbitals of hetero atoms (N, O, S or halogens).
- ✓ They usually occupy the highest energy level of the ground state

Electronic Transitions...



Electronic Transitions...

- $\sigma \rightarrow \sigma^*$ Transitions: An electron in a bonding orbital is excited to the corresponding antibonding orbital.
- The energy required is large because they involve breaking a strong σ bond.
- ✓ For example, methane (which has only C–H bonds, and can only undergo $\sigma \rightarrow \sigma^*$ transitions) shows an absorbance maximum at 125 nm.
- ✓ Absorption maxima due to $\sigma \rightarrow \sigma^*$ transitions are not seen in typical UV-visible spectra (200–700 nm).

Electronic Transitions....

- $n \rightarrow \sigma^*$ Transitions: Saturated compounds containing atoms with lone pairs (non-bonding electrons) are capable of $n \rightarrow \sigma^*$ transitions.
 - These transitions usually need less energy than $\sigma \rightarrow \sigma^*$ transitions.
 - They can be initiated by light whose wavelength is in the range 150–250 nm.
 - The number of organic functional groups with $n \rightarrow \sigma^*$ peaks in the UV region is small.

Electronic Transitions...

- $n \rightarrow \pi^*$ and $\pi \rightarrow \pi^*$ Transitions: Most absorption spectroscopy of organic compounds is based on transitions of n or π electrons to the π^* excited state.
- This is because the absorption peaks for these transitions fall in an experimentally convenient region of the spectrum (200–700 nm).
 - These transitions need an unsaturated group in the molecule to provide the π electron.

Organic compounds containing σ -electrons

- Compounds contain σ -electrons only are the saturated hydrocarbons, which absorb below 170 nm.
- They are transparent in the near UV region (200- 400 nm) and this make them ideal solvents for other compounds studied in this range.
- They are characterized by $\sigma - \sigma^*$ transition only.

Organic compounds containing n-Electrons

- Saturated organic compounds containing hetero atoms, possess n-electrons in addition to sigma-electrons.
- They are characterized by the $s \rightarrow s^*$ and $n \rightarrow s^*$ transitions.
- n-electrons can also be transitioned to π^* when they exist in unsaturated compounds

Organic compounds containing π -Electrons

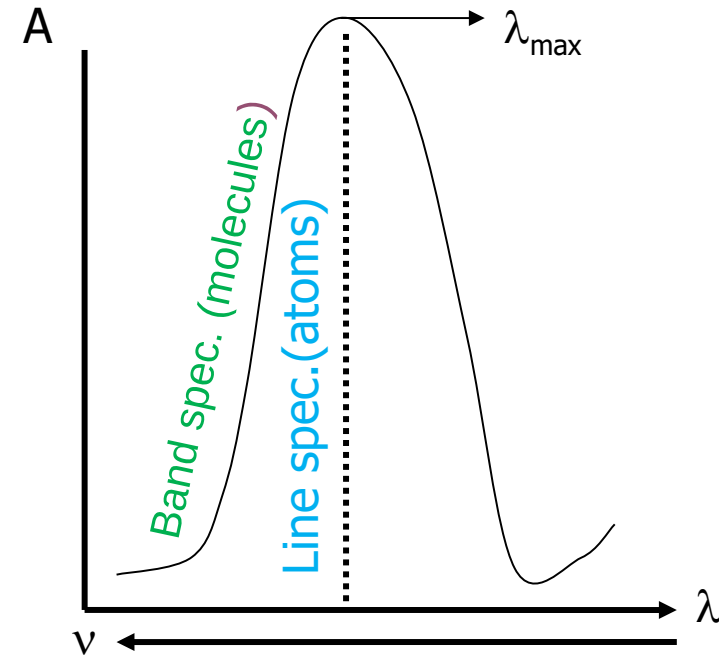
- Unsaturated compounds containing no hetero atoms are characterized by the σ - σ^* and π - π^* transitions, such as ethylene ($\text{CH}_2 = \text{CH}_2$).
- When these compounds containing hetero atoms, they can undergo σ - σ^* , π - π^* , n- π^* and n- σ^* transitions.
- ✓ example acetone ($\text{CH}_3 - \text{COCH}_3$).
- Many inorganic compounds in solution also show absorption in the visible region.
- Increasing order in absorption wavelength

$$\sigma - \sigma^* < \text{n} - \sigma^* < \pi - \pi^* < \text{n} - \pi^*$$

- Of these transitions, the most important are the $n\text{-}\pi^*$ and $\pi\text{-}\pi^*$
- Because they involve functional groups that are characteristic of the analyte and wavelengths that are easily accessible.

Transition	Wavelength range	Examples
$\sigma\text{-}\sigma^*$	<200	C-C, C-H
$n\text{-}\sigma^*$	160-260	H ₂ O, CH ₃ OH, CH ₃ Cl
$\pi\text{-}\pi^*$	200-500	C=C, C=O, C=N, C $\equiv$ C
$n\text{-}\pi^*$	250-600	C=O, C=N, N=N, N=O

- It is the display of the energy level of EMR as a function of wave number or wavelength of EMR energy.
- The energy level of EMR is usually expressed in terms of absorbance, transmission, Intensity.
- It may be:
 - a) **line spectrum**: occurs with atomic spectra such as sodium metal which has a sharp line of λ at 589 nm.



Spectrum...

b) band spectrum: occurs with molecules due to the presence of different vibrational and rotational sub-levels which the molecules may occupy on transition to excited state.

- ***A spectrum tells :***

- ✓ A peak (a valley) represents the $A(T)$ of EMR at that specific wavenumber or wavelength.
- ✓ The wavenumber or λ at the tip of peak is the most important, especially when a peak is broad.

Spectrum...

- ✓ A broad peak may sometimes consist of several peaks partially overlapped each other.
- ✓ The height of a peak corresponds the amount absorption/emission thus can be used as a quantitative information (e.g. conc).
- There are two parameters which define an absorption band :
 1. Its position (λ_{max}) on wavelength scale
 2. Its intensity on the absorbance scale.

Some important terms

Chromophores: (Chrome = color, phore = carrier).

- They are functional groups, which confer color on substances capable of absorbing UV and/or visible light (200 - 700 nm).
- They have unsaturated bonds (double or triple bonds) such as
 $\text{C}=\text{C}$, $\text{C}=\text{O}$, $\text{N}=\text{N}$ and $\text{C}=\text{N}$,etc (π -electrons).

Auxochromes:

- They are functional groups which can not confer colors on substances but have the ability to increase the coloring power of Chromophores.
- They does not absorb radiations longer than 200 nm, but when attached to a given chromophore, causes a shift to a longer wavelength with increase in absorption intensity.
- These are also called colour enhancing groups.

Terms...

Bathochromic (Red) shift:

- Shift of absorption to longer wavelength due to substitution and solvent effects.
- Multiple conjugations

Hypsochromic (Blue) shift:

- It is shift of absorption to shorter wavelength.

Hyperchromic effects:

- It is the increase in absorption intensity respectively.

Hypochromic effects:

- It is the decrease in absorption intensity respectively.

Absorption characteristics of Chromophores

1. Ethylenic Chromophores:

- Their bands are difficult to observe in *near UV* region, so they are not useful analytically.
- However, substitution and certain structural features may cause red shift rendering the band observable in the *near UV* region.

Examples:

- ✓ **Alkyl substitution:** cause red shift due to hyper-conjugation and stabilization of excited state
- ✓ **Exocyclic nature:** cause red shift due to relaxation of strain upon excitation.
- ✓ **Attachment to auxochromes:** cause red shift and increased absorption intensity due to extension of conjugation.

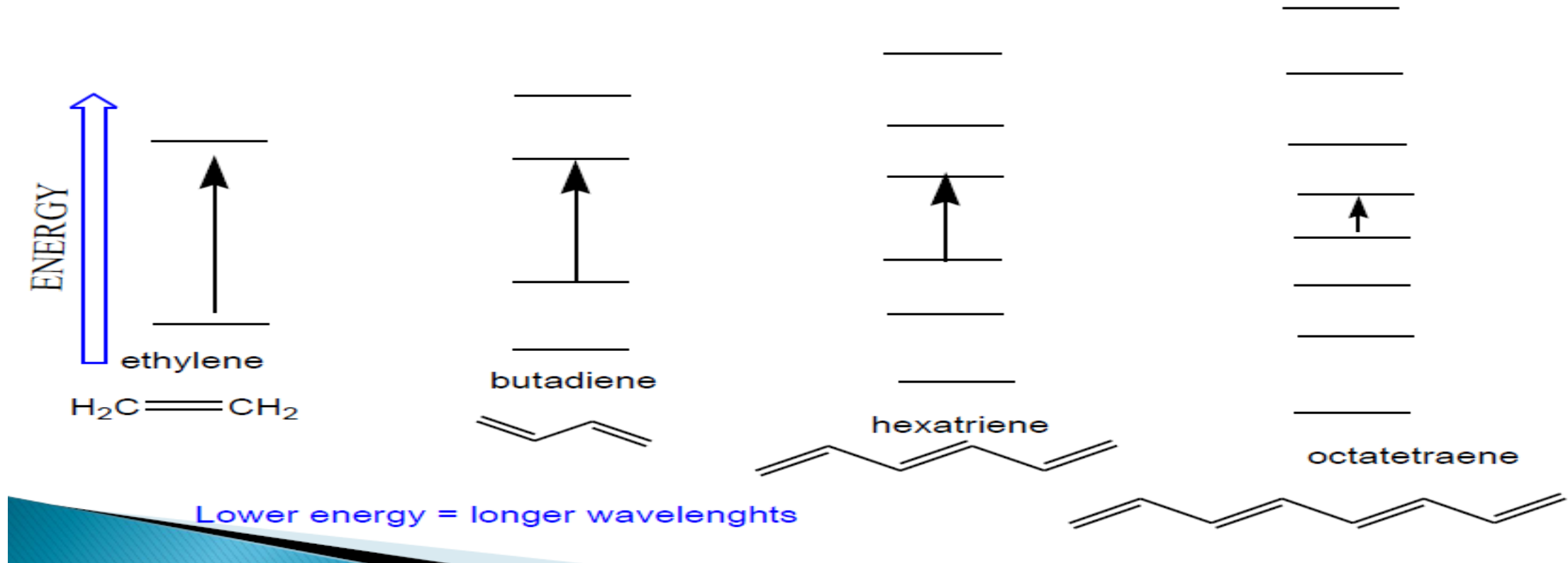
Absorption Characteristics...

- *Carbon-hetero atom chromophores: -C=O, -C=N, -C=S...*
- They exhibit some common characteristics; **n- π^*** band in the
 - range of 275-300 nm., which is the most apparent band, has low
 - intensity and long wavelength.
 - ✓ This band undergoes a **blue shift** on increasing the **solvent polarity** due to increasing the energy of transition as a result of H bonding
 - ✓ **Alkyl substitution**; Cause red shift due to hyper-conjugation.

Absorption Characteristics...

3. Conjugated Chromophores

The more highly conjugated system the smaller the energy change, the lower the frequency and longer the wave length.



Molecule

λ_{max} (nm)

Ethylene

165

1,3-butadiene

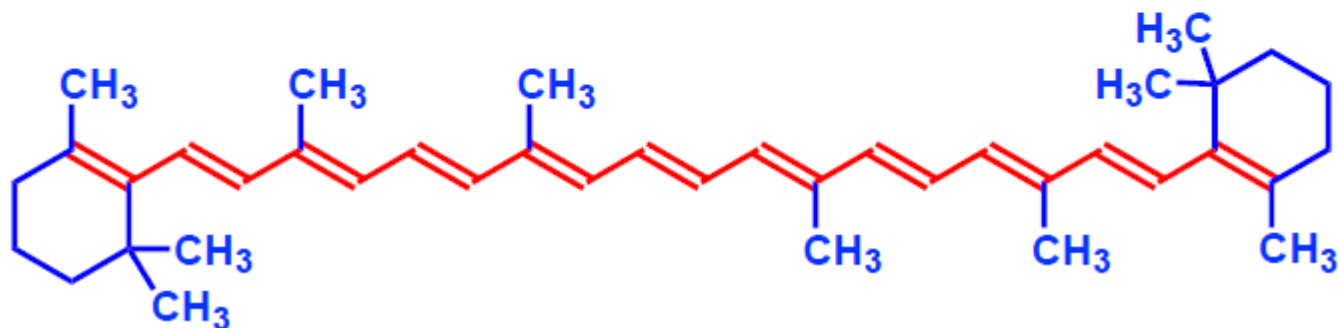
217

1,3,5-hexatriene

258

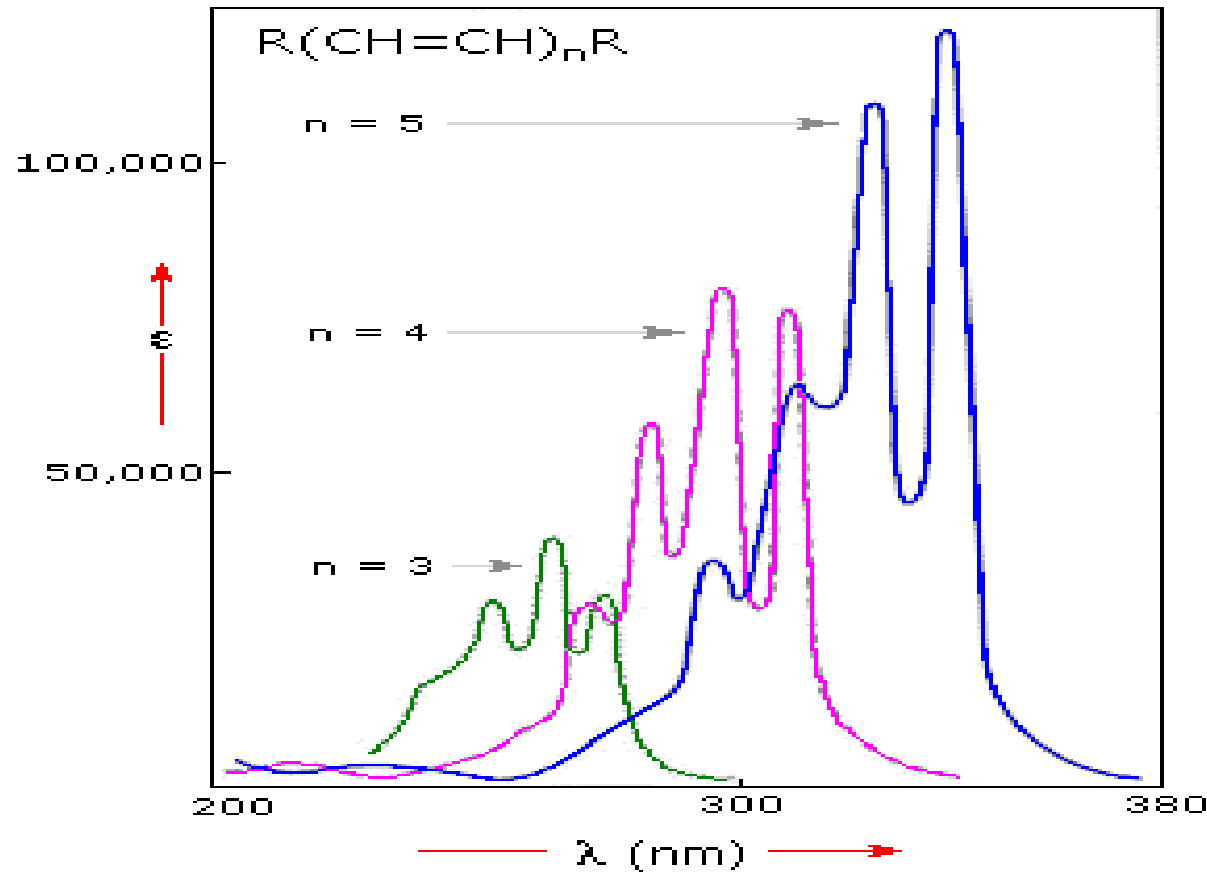
β -Carotene

470



β -carotene, a vitamin found in carrots, and used in food colouring, has eleven conjugated double bonds and its absorption maximum has shifted out of the ultraviolet into the blue region of the visible spectrum, hence it appears bright orange.

The longer the conjugated carbon chain in the absorbing system, the greater the intensity of the absorption. This is shown by the spectra of the polyenes $\text{CH}_3-(\text{CH}=\text{CH})_n-\text{CH}_3$, where $n=3,4$ and 5

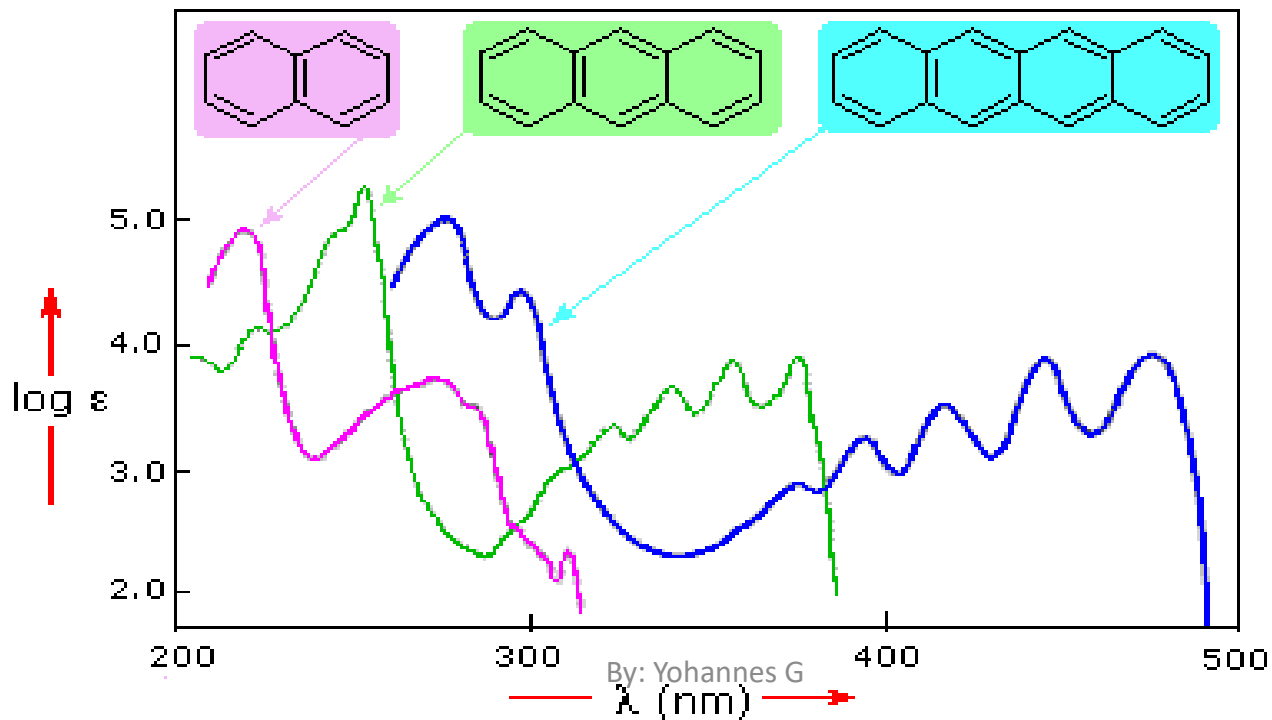


Ultraviolet/visible spectra of the polyenes CH_3-
 $(\text{CH}=\text{CH})_n-\text{CH}_3$

By: Yohannes G
 where $n=3,4$ & 5

Added conjugation of benzene

- ▶ The added conjugation in naphthalene, anthracene and tetracene - bathochromic shifts of absorption bands.
- ▶ All the absorptions do not shift by the same amount
for anthracene and tetracene the weak absorption is obscured by stronger bands that have experienced a greater red shift.
- ▶ As might be expected from their spectra, naphthalene and anthracene are colorless, but tetracene is orange.



Effect of pH on absorption spectra

- Changes in PH primarily affect molecules containing functional groups that can gain or lose proton (H^+) depending on the surrounding acidity/basicity.

✓ Ionizable groups

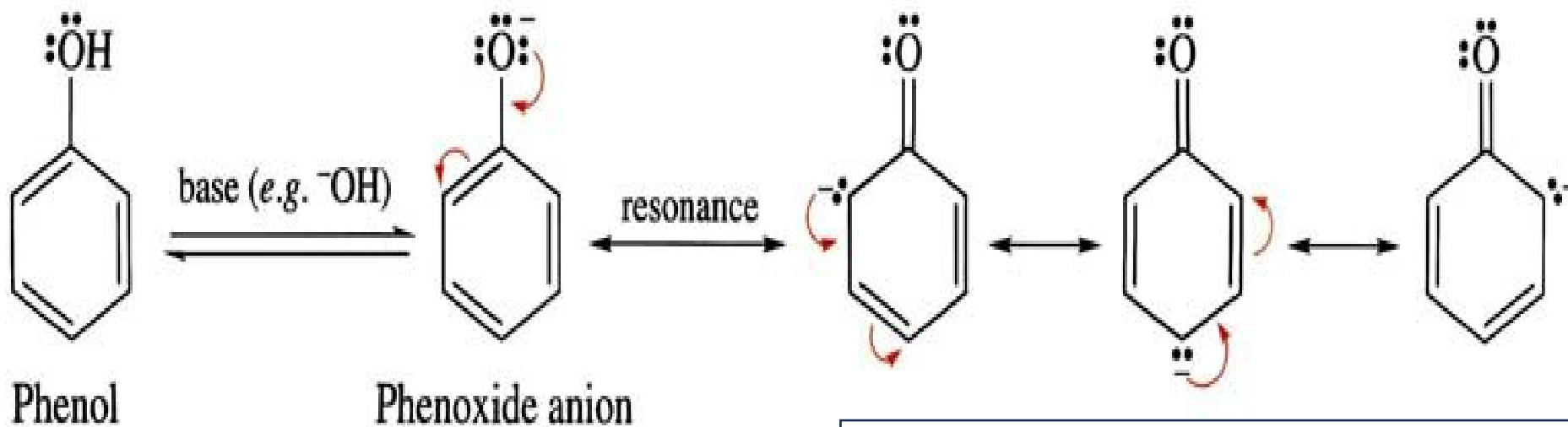
Common ionizable groups:

- Carboxylic acids ($COOH$): can lose H^+ in basic conditions.
- Amines (NR_2): can gain H^+ in acidic conditions.
- Phenols ($ArOH$): can lose H^+ in basic conditions

Effect of pH

- Spectrum in alkaline medium exhibits **bathochromic shift** with **hyperchromic effect**.
- The red shift is due to the participation of the pair of electrons in resonance with the π electrons of the benzene ring, thus increasing the delocalization of the π electrons.

Effect of pH upon the absorption maxima of phenols



Neutral or acidic conditions

λ_{max} (H_2O) 210 (6000)
270 (1500)

Basic conditions

λ_{max} (H_2O) 235 (9400)
287 (2600)

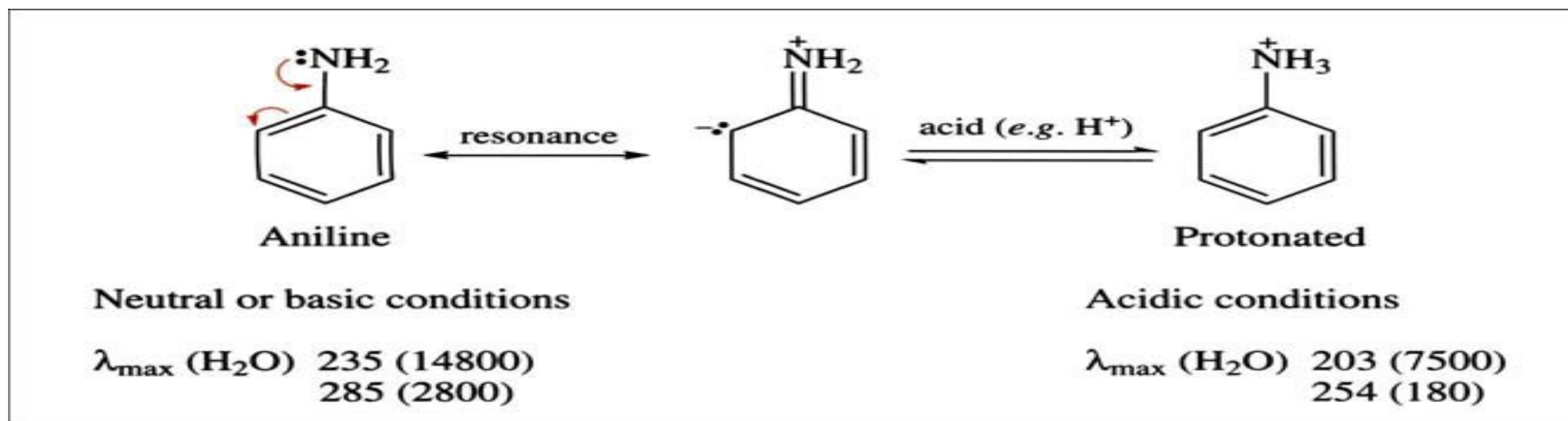
- ✓ In Phenoxide ion, lone pair on oxygen delocalized the π system of the aromatic ring.
- ✓ Conjugation increases and leads to decrease in energy gap.
- ❖ Bathochromic shift and increase in intensity (hyperchromic effect?)

Effect of pH upon the absorption maxima of an aromatic amine

Treatment of an aromatic amine with acid, causes **protonation** of the amine

➤ leads to a loss of the overlap between the amine lone pair and the aromatic π -system.

The result is a **blue or hypsochromic shift** (to shorter wavelength) along with a decrease in intensity.



Effects of Solvent on Absorption spectra

- Different Solvents can cause shifts in the absorption bands due to their varying **polarity and interactions with the solute**.

❖ Factors influencing Absorptions

Solvent-Solute Interactions: affect the energy levels of the electrons in the solute.

- This can lead to shifts in the absorption bands of the solute.

Types of Interactions: hydrogen bonding, dipole-dipole interactions, and London dispersion forces.

- ✓ These interactions can either stabilize the ground state or the excited state of the solute, influencing the energy difference between them.

Effects of Solvent....

Types of Shifts:

A. Bathochromic Shift: When the solvent interaction lowers the energy gap between the ground and excited states of the solute, the absorption band moves towards longer wavelengths (lower energy) in the spectrum.

- Polar solvents are more likely to cause bathochromic shifts due to their stronger interactions with solutes.

Effects of Solvent...

- Polar solvents stabilize polar groups through dipole-dipole interaction or hydrogen bonding.
 - ✓ If the excited state of a polar molecule has a higher dipole moment than its ground state (the excited state is stabilized by polar solvents to a greater extent than the ground state)
 - Thus there is a bathochromic shift with increasing solvent polarity.
 - ✓ If chromophore is more polar in the ground state than in the excited state, then polar solvents stabilize the ground state to greater extent than the excited state. (in few cases)
 - Thus there is a hypsochromic shift with increasing solvent polarity

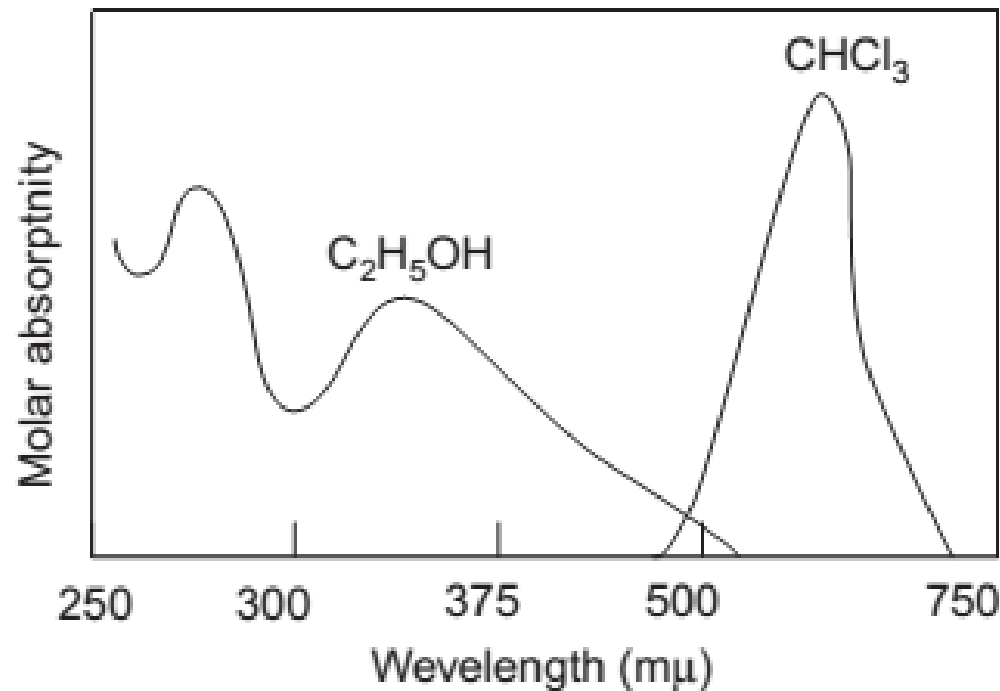
Effects of Solvent

B. Hypsochromic Shift: Conversely, if the solvent interaction increases the energy gap between the ground and excited states, the absorption band moves towards **shorter wavelengths (higher energy)** in the spectrum.

Nonpolar solvents are more likely to cause hypsochromic shifts due to weaker interactions with solutes.

Effects of Solvent...

- The nonbonding electrons of nitrogen & oxygen usually interact with polar solvents that ultimately give rise to a characteristic shift to **shorter wavelengths**.



Absorption spectrum of Iodine in polar & nonpolar solvents

Basic Requirements

- 1. Transparency in the UV range:** The solvent shouldn't absorb UV light itself in the wavelength range of interest.
 - ✓ Ideally, the solvent should have a cut-off wavelength below the region you want to analyze.
- 2. Dissolves the Sample:** The solvent needs to be able to dissolve a sufficient amount of your sample for an accurate measurement.
- 3. Chemical Inertness:** The solvent shouldn't react with the analyte or the cuvette (the container that holds the sample) during the measurement.

Basic Requirements...

4. Low Volatility: A solvent with low volatility minimizes evaporation during the measurement process.

✓ Evaporation can concentrate the sample and affect the measured absorbance.

5. Suitable Refractive Index (for quantitative analysis): In quantitative analysis, matching the refractive index of the solvent to the cuvette material can minimize reflection losses that could affect the measurement.

▪ However, this is less critical for qualitative analysis where you're just identifying the presence of a compound.

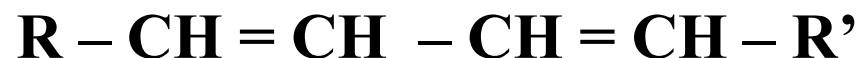
Calculation of λ_{max} of an organic compound

Woodward-Fieser Rules

- ❖ It is used to calculate the position and λ_{max} for a given structure by relating the position and degree of chromophore.

Conjugated Dienes

- Each type of diene or triene system is having a certain fixed value at which absorption takes place ; this constitutes the base value.
- These rules specify a base value (**214 nm**) for the parent diene which is 1,3-butadiene.

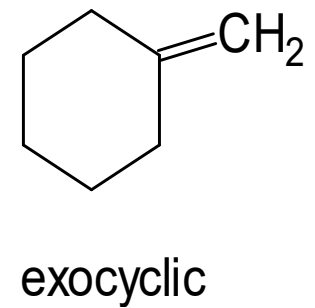
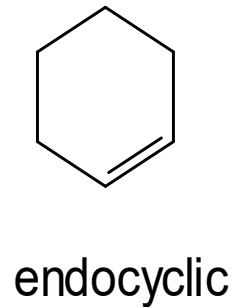
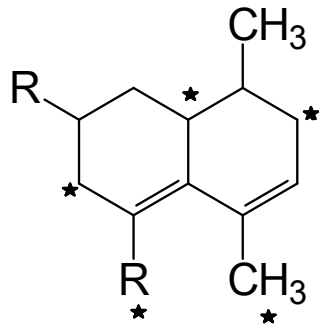


- are found in most organic compounds
- According to woodward's rules the λ_{max} of the molecule can be calculated using a formula:

$$\lambda_{\text{max}} = \text{Base Value} + \sum \text{substituent contributions} + \sum \text{other contributions}$$

Calculation of λ_{max}

- The value of 214 nm is red-shifted upon:
 - alkyl substitution or attachment of ring carbons (Ring residues)
 - by the presence of double bonds outside (**exocyclic**) a ring.



- ★ Counted Ring residues and alkyl substitutions

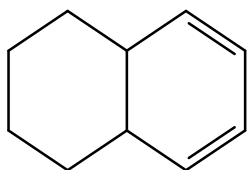
Calculation of λ_{max}

- The first step in predicting the wavelength of maximum UV absorption for conjugated dienes is to determine whether it lies in a s-trans or s-cis conformation.
- If it lies in the s-trans conformation its base wavelength is 217nm.
- If it lies in the s-cis conformation, its base wavelength is 253nm

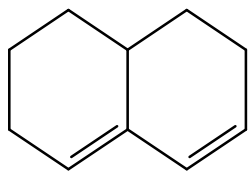


Calculation of λ_{max}

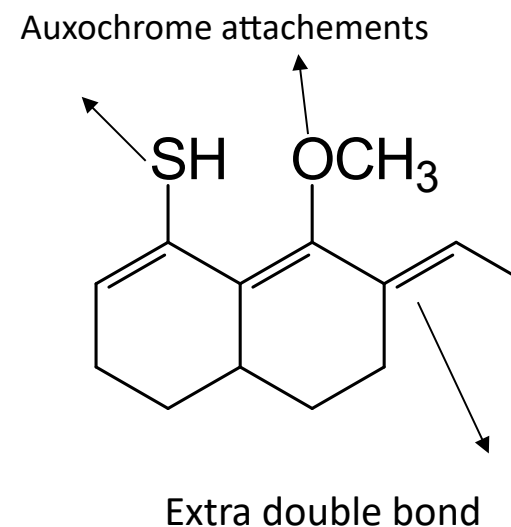
- Also the presence of the two double bonds within the same ring (**homoannular**) gives a different value from that of **heteroannular** systems where the two double bonds occur in two different rings (and still in conjugation).
- Addition of **EXTRA** double bonds in conjugation.
- Attachment of **AUXOCHROMES** to the conjugated system.



homoannular



heteroannular



Base values:

- | | |
|---|--------|
| ➤ Base value for an unsubstituted, conjugated, acyclic or heteroannular diene | 214 nm |
| ➤ Base value for an unsubstituted, conjugated, homoannular diene | 253 nm |

Increments for:

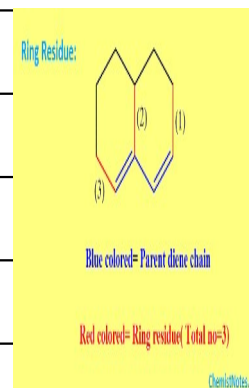
- | | |
|--|---------|
| Each extra double bonds in conjugation | + 30 nm |
| Exocyclic double bond (effect is two fold if the bond is exocyclic to two rings) | + 5 nm |

Substituent effect:

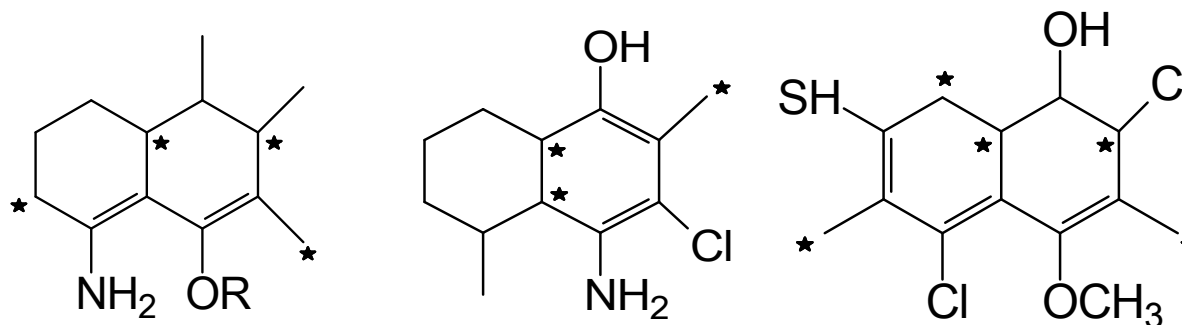
- | | |
|--|---------|
| A. -OCOR or -OCOAr | + 0 nm |
| B. Simple alkyl substituents or ring residue | + 5 nm |
| C. Halogen (-Cl, -Br) | + 5 nm |
| D. OR (R=Alkyl) | + 6 nm |
| E. SR (R=Alkyl) | + 30 nm |
| F. NR ₂ (R=Alkyl) | + 60 nm |

Woodward Rules for Conjugated Dienes can be summarized as :

Component	nm
Base value for heteroannular or opened-chain dienes	214
Base value for homoannular dienes	253
Add the following Values to the base value:	
(a) Each extra double bond in conjugation	30
(b) Each Alkyl Substituent or ring residue	5
(c) Each exocyclic nature	5
(d) Each auxochrome has its corresponding value:	
- OAc	0
- OR (including OH)	6
- SR (including SH)	30
- Cl or Br	5
- NR ₂ (including NH ₂ & NHR)	60
(e) Solvent Correction	0



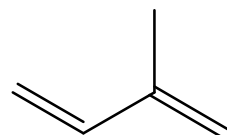
The following examples illustrate the use of these rules:



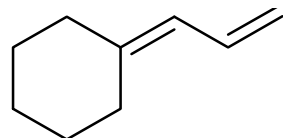
Basic Value	214	253	253
Extra D.B.	---	---	30
Exocyclic D.B.	5	---	5
Ring residue	15	10	15
Alkyl Substituent	5	5	10
Auxochromes			
OR	6	6	6
SR	---	---	30
Cl & Br	---	5	5
NR_2	60	60	---
Calculated λ_{max}	305	339	354

Woodward-Fieser Rules - Dienes

For example:

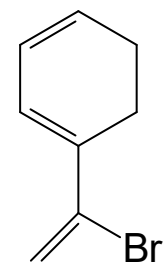
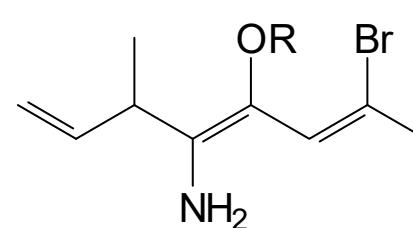
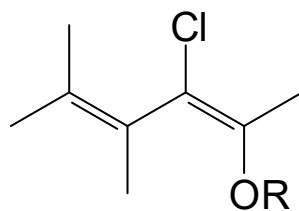
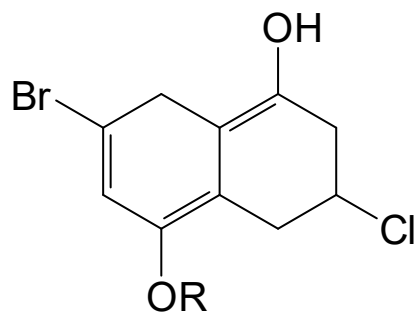
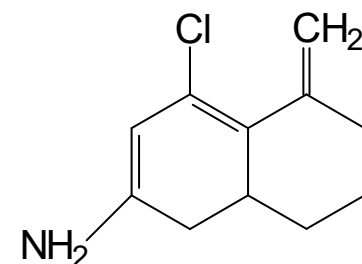
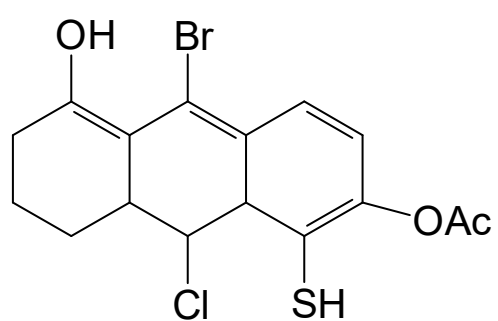
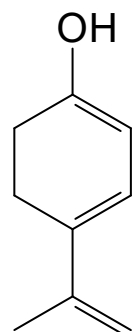
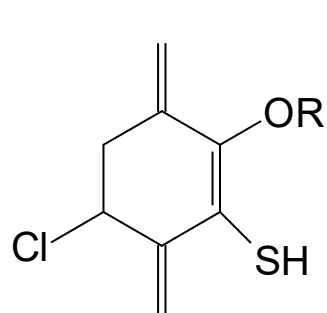
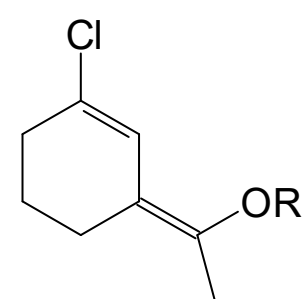
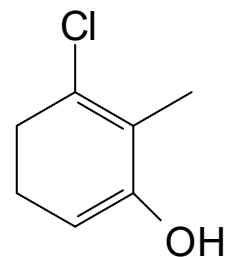
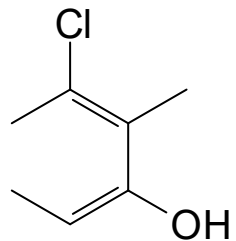
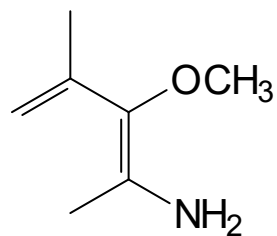


Isoprene - acyclic butadiene =	214 nm
one alkyl subs.	<u>+ 5 nm</u>
	219 nm
Experimental value	220 nm



Allylidencyclohexane	
- acyclic butadiene =	214 nm
one exocyclic C=C	+ 5 nm
2 alkyl subs.	<u>+10 nm</u>
	229 nm
Experimental value	237 nm

Calculate the λ_{max} of the following compounds :



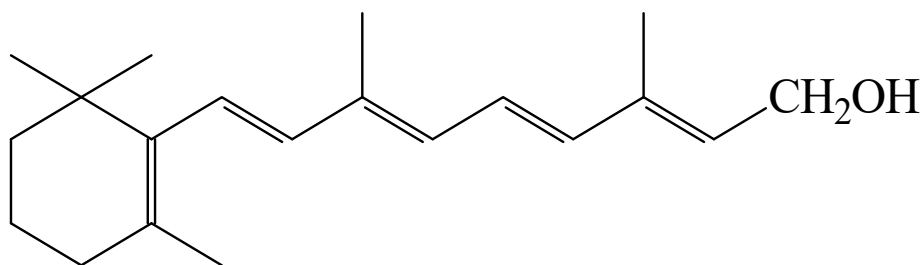
II- Simplified Kuhn and Hausser rules:

These rules can be used for calculating $\lambda_{\max}$ for conjugated polyenes as follows:

$$\lambda_{\max} (\text{nm}) = 134 \sqrt{n} + 31$$

where n is the number of conjugated double bonds.

Example : Calculate the $\lambda_{\max}$ of the following compound :



$$\lambda_{\max} = 134 \sqrt{5} + 31 = 330.6 \text{ or } 331 \text{ nm}$$

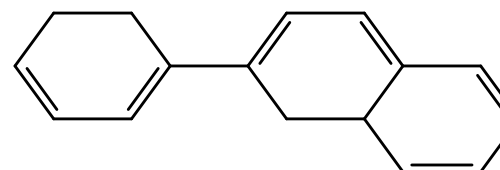
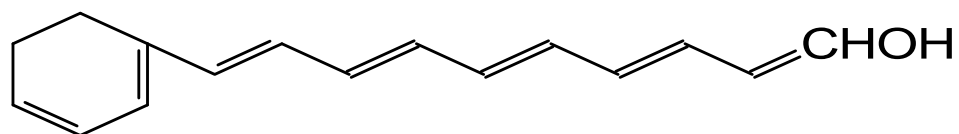
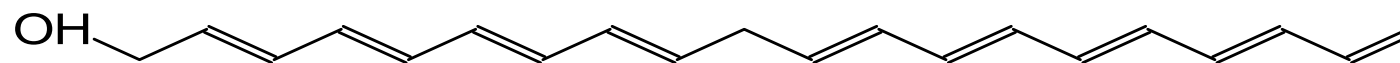
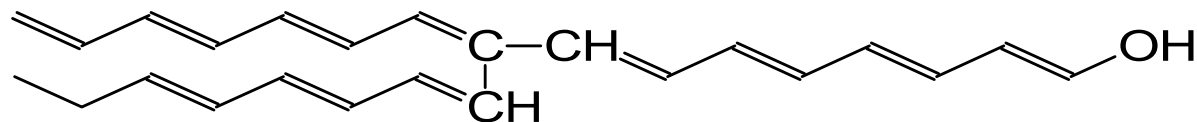
This rule is also useful for calculating number of double bonds from the observed $\lambda_{\max}$ as $n = (\lambda_{\max} - 31/134)^2$

Example : If $\lambda_{\max}$ of a compound is 433 nm calculate the approximate number of double bonds :

$$\text{The number of double bonds } (n) = [(433 - 31) / 134]^2 = 9$$

➤ Using the simplified Kuhn and Hausser rules, Calculate the approximate

$\lambda_{\max}$ for the following compounds :



Calculate the approximate number of double bonds present in each compound , if you gave the following $\lambda_{\max}$ for each:

1- 420 nm ,

2- 530 nm ,

3- 485 nm

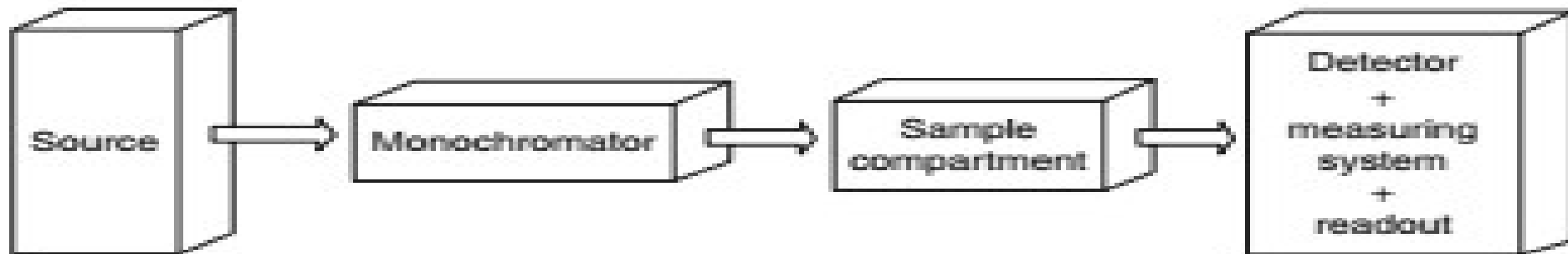
4- 565 nm

5- 612 nm

6- 710 nm

Instrumentation

- The different components are the following:
- (a) Radiation source
- (b) Monochromators
- (c) Sample cells
- (d) Detector
- (e) Recorder (or) display



Flow chart of instrument of UV spectroscopy

Light Sources. Two types:

1- ***Continuous Sources***: which produce spectra over a broad range

- ✓ Tungsten lamp (provides visible spectrum; 400-1200 nm)
- ✓ Deuterium lamp (provides ultra-violet spectrum; 190-400 nm)

2- ***Discontinuous or Discrete Sources***: which produce only specific (discrete) wavelengths .

- ✓ Hollow cathode lamp (HCL)
- ✓ Electrodeless discharge lamp (EDL)

Radiation Source

1. Tungsten – Halogen lamp: This lamp covers the wavelength ranging from the red end of the visible spectrum (750–800 nm) to the near ultraviolet (300–320 nm).

✓ The lamp is provided with a quartz outer sheath to permit the use of the ultraviolet part of the emission.

2. Hydrogen (or) deuterium lamp: This is mainly used for the measurement of far ultraviolet (down to 200 nm).

✓ It consists of two electrodes dipped in a deuterium filled silica envelope.

Light source...

For visible region

- The visible region spectrum ranges from 370 to 750nm
- Hence any lamp source w/c gives adequate intensity of radiation over the entire wavelength region can be used.
 - Tungsten filament lamp - is a satisfactory light source.
 - Carbon arc lamp

➤ UV region

- The best source of UV:
 - More stable
 - more intense &
 - Gives range of spectrum from 180-360nm(upto400nm)

1. Hydrogen discharge lamp/HDL

- Gives radiation from 120-350
- Consists of hydrogen under high pressure

2. Deuterium lamp

- It consist of two electrodes contained in a Deuterium filled silica envelop.
- Give a continuous spectrum in the range 185-380nm.
- Offers 3-5 times more intensity than other types.
- Most widely used, but expensive

3. Xenon discharge lamp

- Filled with xenon at 10-30 atmospheric pressure
- Offers greater intensity than HDL

4. Mercury arc lamp

- Gives sharp bands/the spectrum is not continuous
- Hence it is not widely used.

Filters & Monochromators

➤ wavelength selectors

(polychromatic light into monochromatic light)

❖ Filters -two kinds:

Absorption & Interference filters

❖ Monochromators - two kinds:

- Prism type-dispersive type & littrow type
- Grating type-diffraction grating & transmission grating

It consists of three parts: an entrance slit, a dispersing element, and an exit slit.

- ◆ Entrance slit-used to create a unidirectional beam of light of appropriate intensity
- ◆ Dispersing element-disperse the light into its component wavelength.
- ◆ The Monochromatic light is then selected by an exit slit

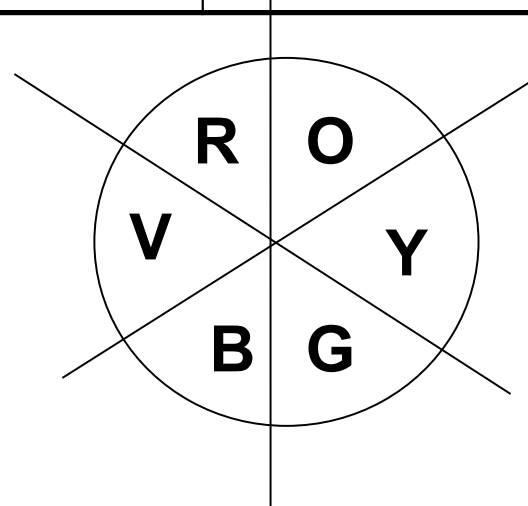
Filters:

I. Absorption filter

- Made up of glass, coated with pigments /colored glass or
- Made up of dyed gelatin between two glass plates
- These filters can be selected according to the procedure given below:
 1. Draw a filter wheel (circle with 6 parts)
 2. write the color (VIBGYOR) in anticlockwise manner, omitting indigo
 3. If the color of the solution is **RED**, use **GREEN** filter
- The color absorbed is the complement of the color of the filter.

Wavelength (nm)	Color Absorbed	Color Transmitted
350-430	Violet	Yellow
430-475	Blue	Orange
505-555	Green	Red
575-600	Yellow	Violet
600-650	Orange	Blue
650-700	Red	Green

V
~~**I**~~
B
G
Y
O
R



Advantages

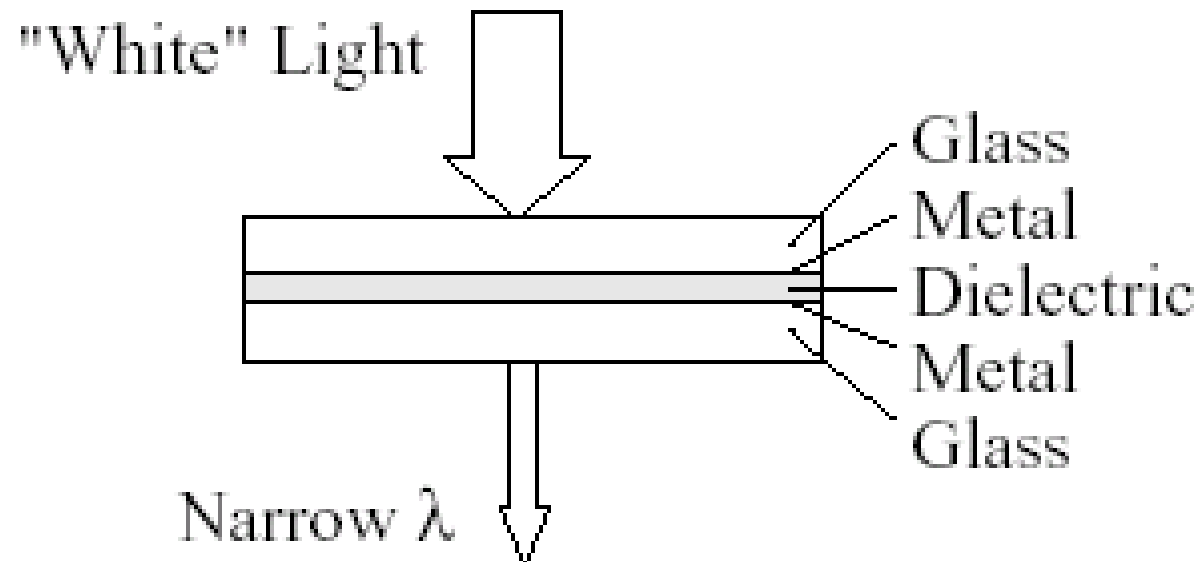
- Simple in construction
- Cheaper
- Selection of filter is easy

Disadvantages

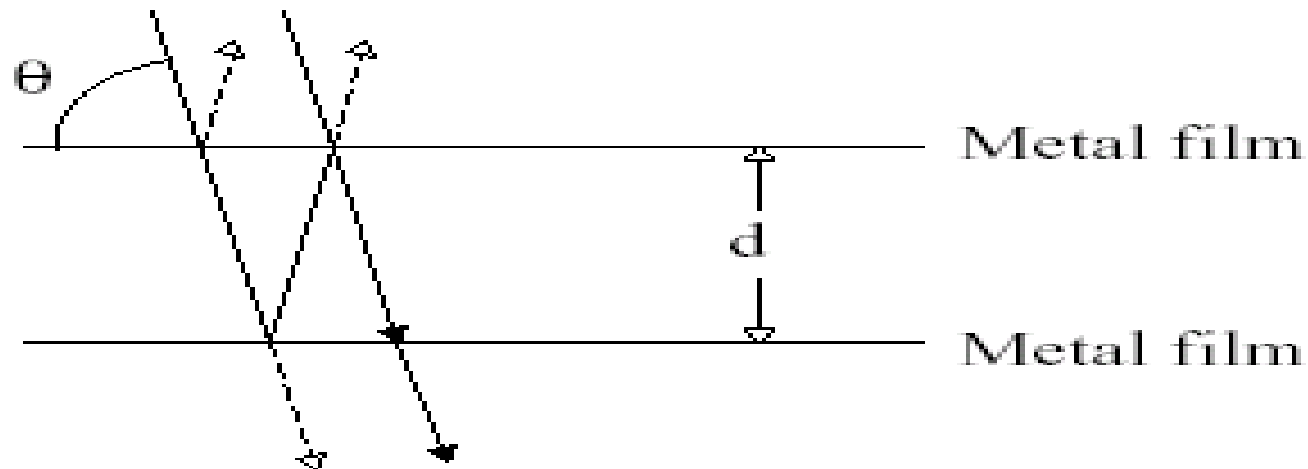
- Less accurate
- Band pass is more($\pm 20-30$)
- Intensity of radiation becomes lessened due to absorption by filters.

II. Interference (Fabry-Perot) filter

- Two thin sheets of metal sandwiched between glass plates, separated by transparent material



- The dielectric spacer film made up of CaF_2 , MgF_2 or SiO_2 , b/n two parallel reflecting silver films
- The radiation reflected by the 2nd film & the incoming radiation undergoes constructive interference to give monochromatic radiation.
- Interference for transmitted wave through 1st layer and reflected from 2nd layer



Advantages

- Inexpensive
- Has lower band pass than that of absorption filter, (10-15nm Vs 20-30nm) 

More accurate

E.g. To measure at 400nm, we get radiation from 385-415nm(IF) & 370 - 430nm(AF)

- Use of additional filter cuts off undesired λ

Disadvantages

- Peak transmission is low when additional filters are used
- The band pass is only 10-15nm & hence higher resolution, like with monochromators, cannot be achieved.

Monochromators:

- Entrance slit -to get narrow source
- Collimating lens or mirror -to render light parallel
- Dispersion element (prism or grating)-to disperse radiation into its component wavelength
- Focusing lens or mirror-to reform the images of entrance slit
- Exit slit -to fall on sample cell

Prism

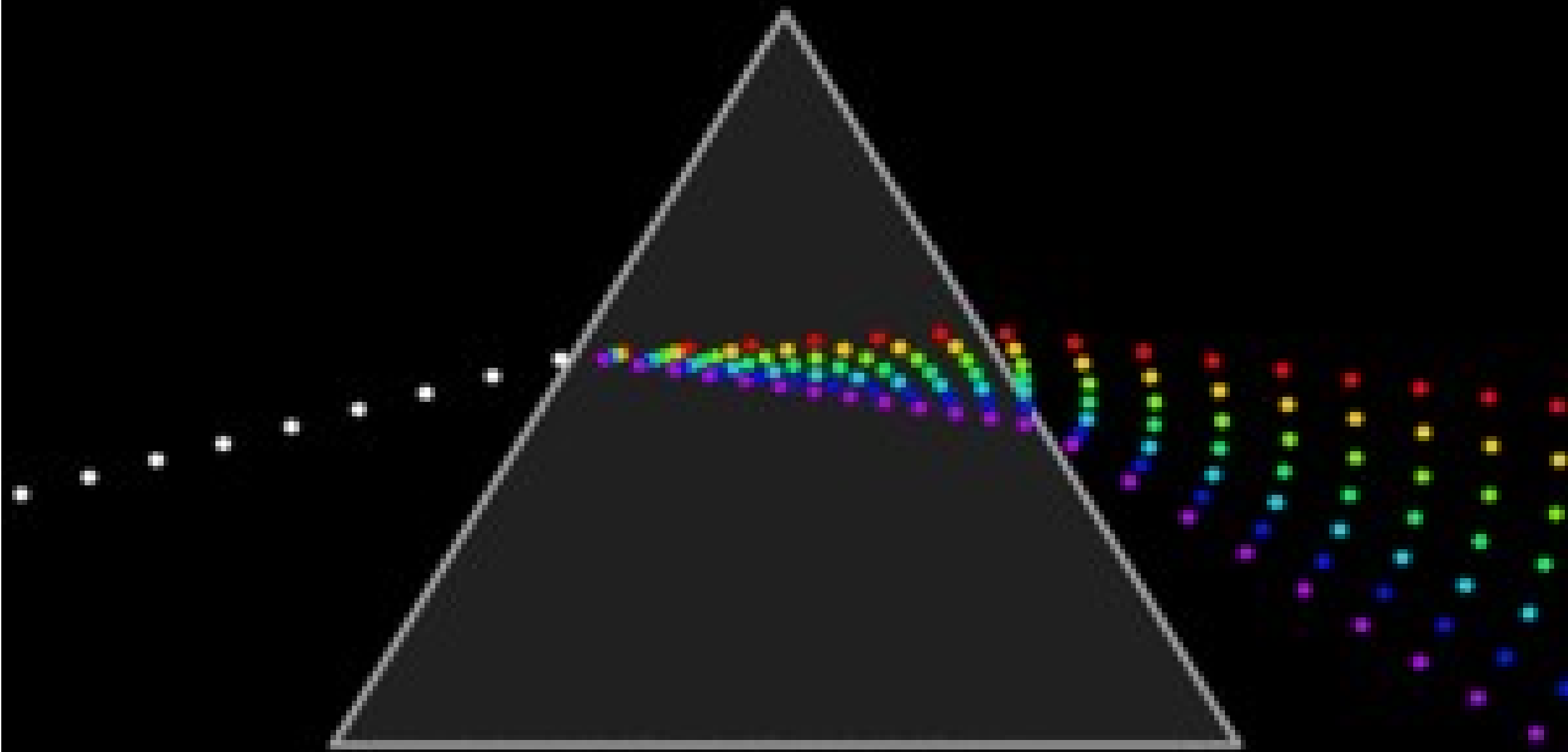


Grating



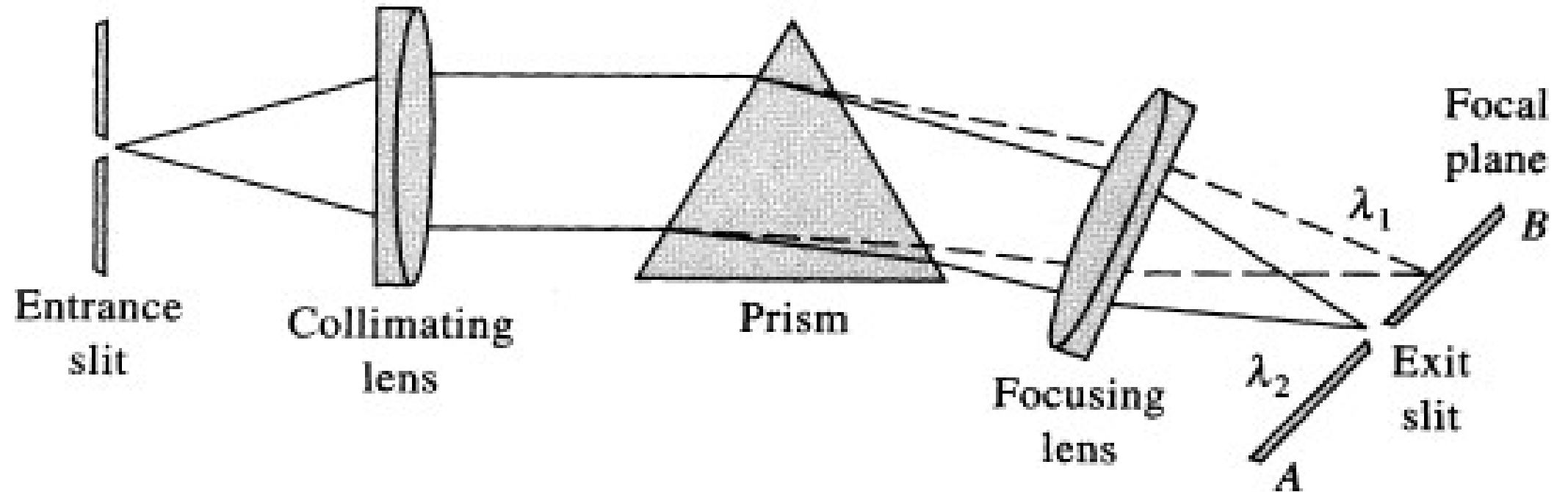
Prism

- Has lower band pass than that of filters ➡ better resolution
- The resolution depends upon the size & refractive index of the prism.
- Two types of prisms:
 - dispersive type &
 - Littrow type

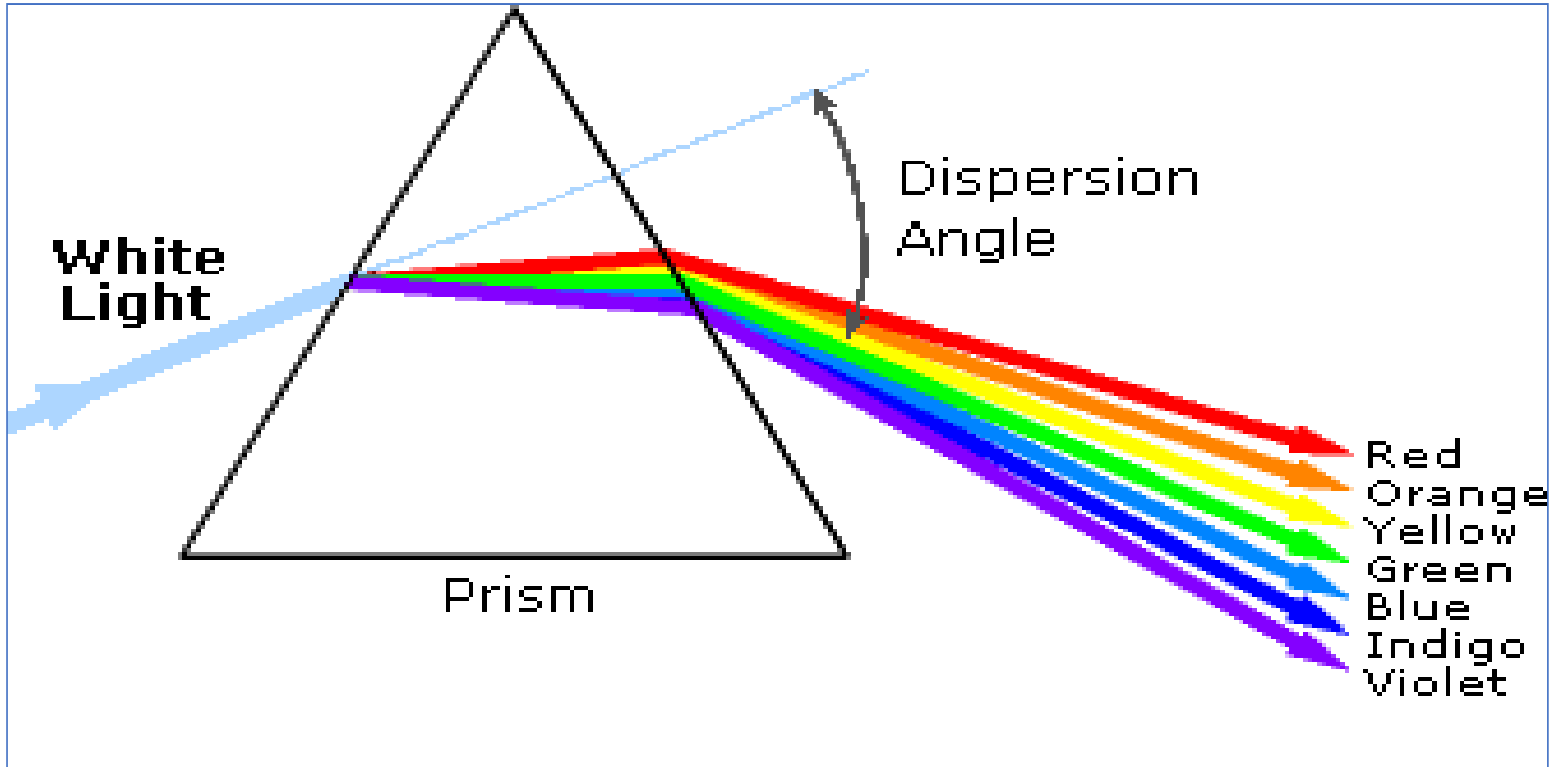


I. Refractive type – Dispersive type

Typical Prism Monochromator (Fig 7-16)



I. Refractive type – Dispersive type...



- The required radiation on the exit slit can be selected by:
 - rotating the prism or
 - keeping the prism stationary & moving the exit slit.

II. Reflective type – Littrow type mounting

- The principle of working is similar to the reflective type except that, a reflective surface is present on one side of the prism.
- Hence the dispersed radiation gets reflected & can be collected on the same side as the source of light.

Grating

- The most efficient in converting a polychromatic light in a real sense.
- Because the resolution of $\pm 0.1\text{nm}$ could be achieved.
- Two types of grating:

I. Diffraction Grating (most modern instruments)

- Gratings are rulings made on some materials like glass, quartz or alkyl halides, depending on the instrument, whether it is UV/Vis/IR spectrophotometer, respectively.
- The number of rulings are ranges from 20lines/mm(IR) to 3600grooves or more per mm for UV/Vis spectrophotometer

Diffraction Grating ...

- To make the surface reflective, a deposit of aluminium is made on the surface.
- The rays which are incident upon the grating get reinforced with the reflected rays &
- Hence the resulting radiation has wavelength which is governed by the equation:

$$m\lambda = b(\sin i + \sin r)$$

Where λ = wavelength of light produced

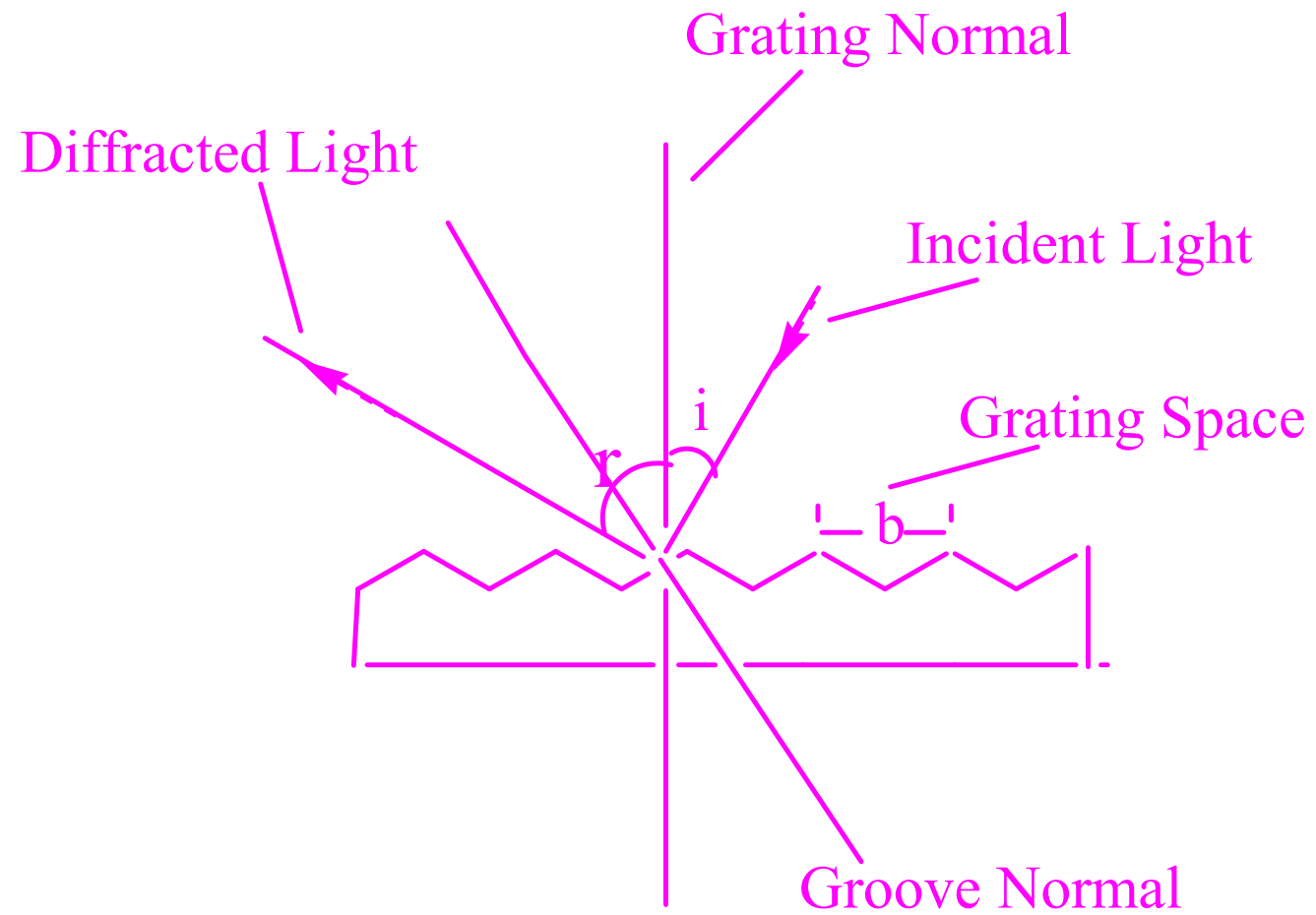
b = grating space

i = angle of incident light

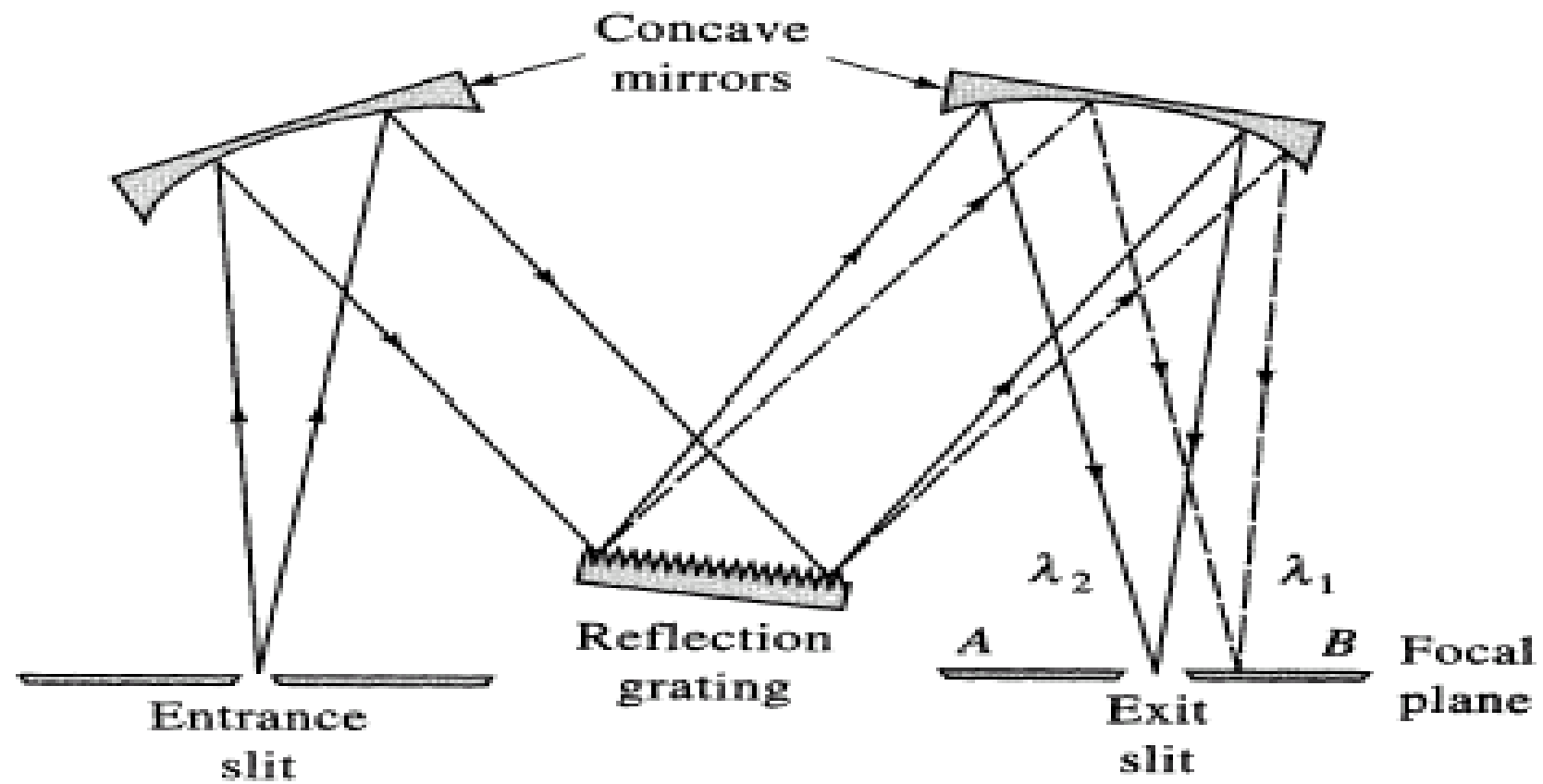
r = angle of reflection

m = order No. (0, 1, 2, 3, etc)

- Band pass $\pm 0.1 \text{ nm}$



Grating Monochromator (Fig 7-17)



II. Transmission Grating

- Similar to diffraction grating, but refraction takes place instead of reflection.
- The radiation transmitted through grating reinforces with the partially reflected radiation.
- The wavelength of radiation produced will be:

$$\lambda = d \sin \theta / m$$

Where $d = 1/\text{lines per cm}$

θ = angle of deflection

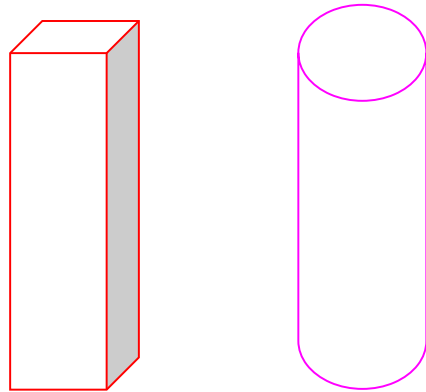
II. Transmission Grating...

- Thus a light radiation at any angle(θ) or any order can be collected by either:
 - moving the grating & fixing the slit or
 - moving the slit & keeping the grating constant.
- The mirrors, grating etc should be made up of quartz for UV spectrophotometer.

Sample holder/ compartment

- The container is called cuvette.
- The material of cuvette should not absorb at the wavelength being observed .
- For the UV region sample compartment is made of quartz since quartz will not absorb in the UV region.
- For the Visible region, compartment composed of simple glass since glass will absorb in the UV but not absorb in the visible region.
 - ◆ In UV any solvent that does not have any $\tilde{N}$ -bonding can be used including water.
 - ◆ In the visible region any solvent that is colourless can be used including water.

Shape of cell: Cylindrical or rectangular



Visible = glass



UV = quartz

Detectors

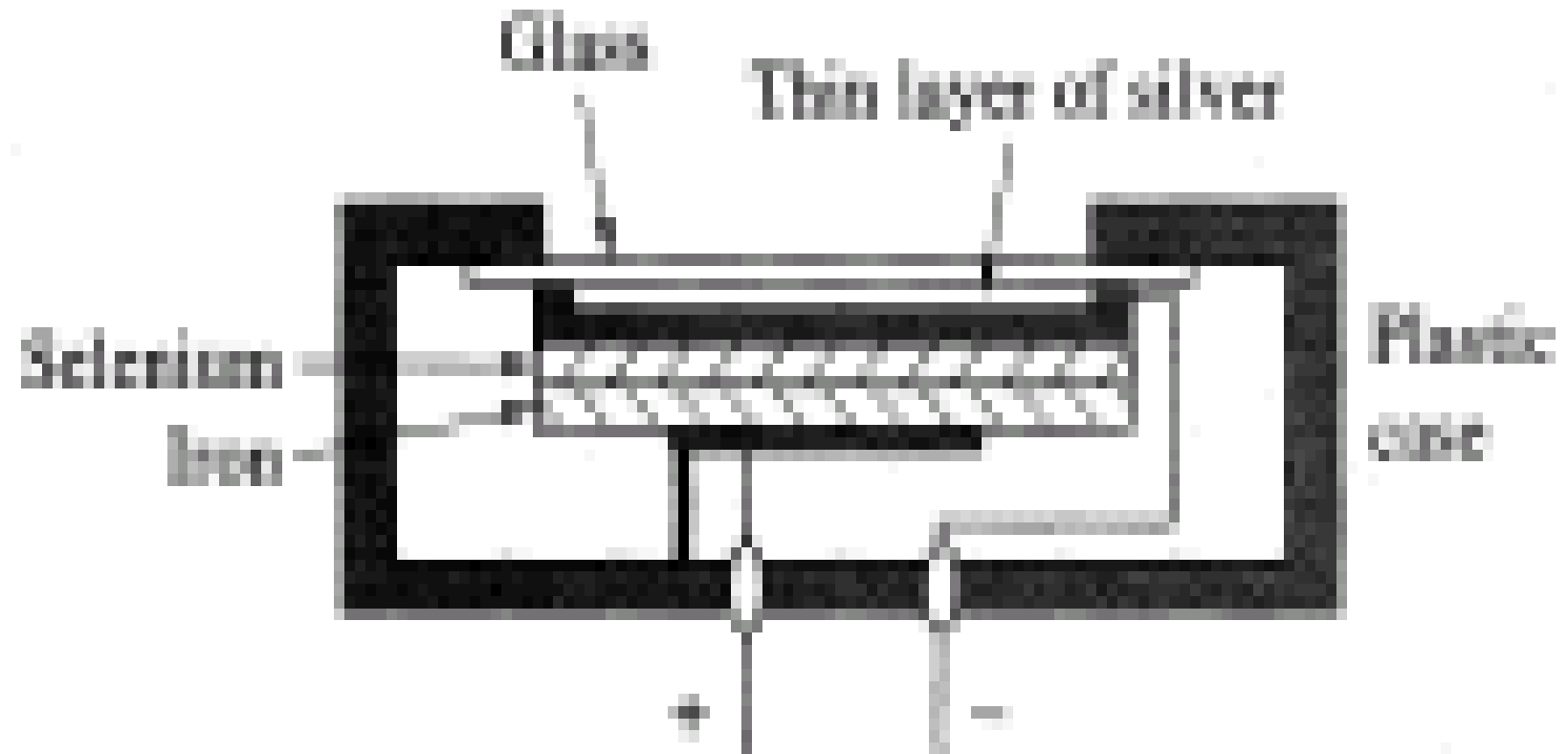
- Two Types of detectors are used in this respect:
 - 1- Heat Sensitive Detectors
 - 2- Photoelectric Detectors
- ✓ Photoelectric detectors are the most frequently used for this purpose.
- ✓ They give electrical signal, which is directly proportional to the intensity of the transmitted light.

Photodetectors

- Used to convert transmitted light energy into an equal amount of electrical energy.
- The most commonly used detectors are:
 - Barrier-layer cell
 - Phototube
 - Photomultiplier tube

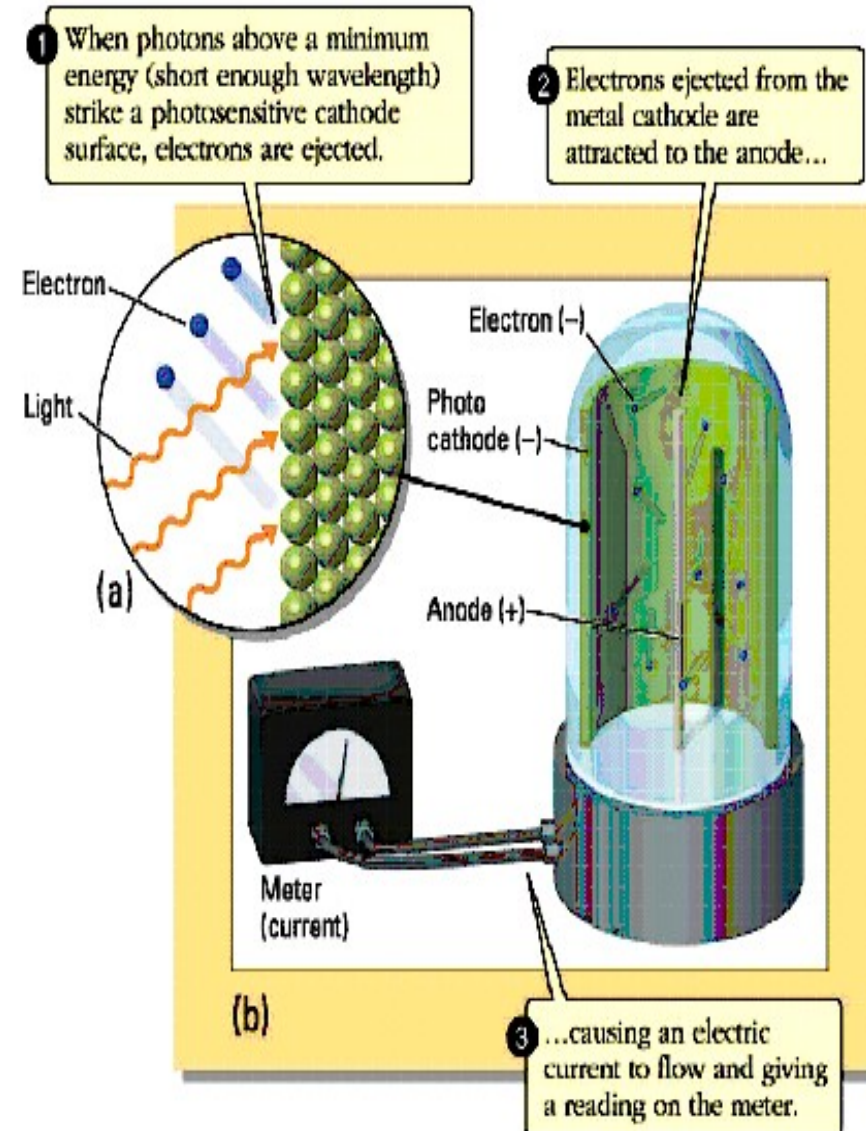
A) Photovoltaic cells – Barrier layer cells

- metal-semiconductor-metal sandwiches that produce voltage when irradiated (350-750 nm)
- Used in inexpensive instruments, like filter type colorimeters.



(A) Photocells (Phototubes):

- Light (radiant energy) falls on the cathode surface which excites electrons and generates an electric current which is proportional to light intensity
- (In other words) Converts the energy of an incoming photon into a current pulse. Conversion is done on a photoemissive surface by the “photoelectric effect”

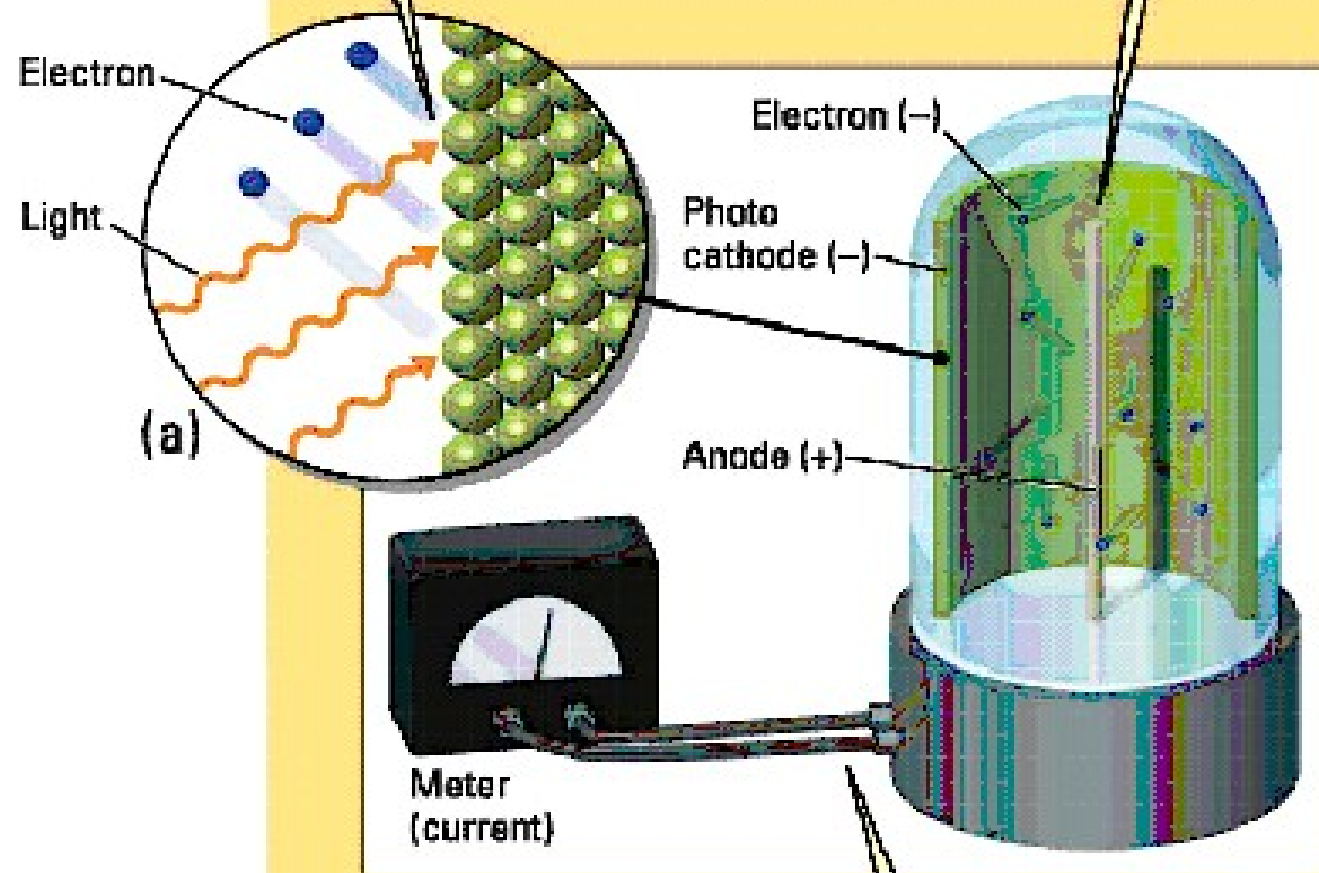


B) Phototube - Photo Emissive cells

- electrons produced by irradiation of cathode travel to anode.
- response depends on cathode material (200-1000 nm)
- The photocathode is coated with elements of high atomic volume like Caesium, potassium or silver oxide, which can liberate electrons, when light radiation falls on it.
- It has better sensitivity as compared to photovoltaic cell.

1 When photons above a minimum energy (short enough wavelength) strike a photosensitive cathode surface, electrons are ejected.

2 Electrons ejected from the metal cathode are attracted to the anode...

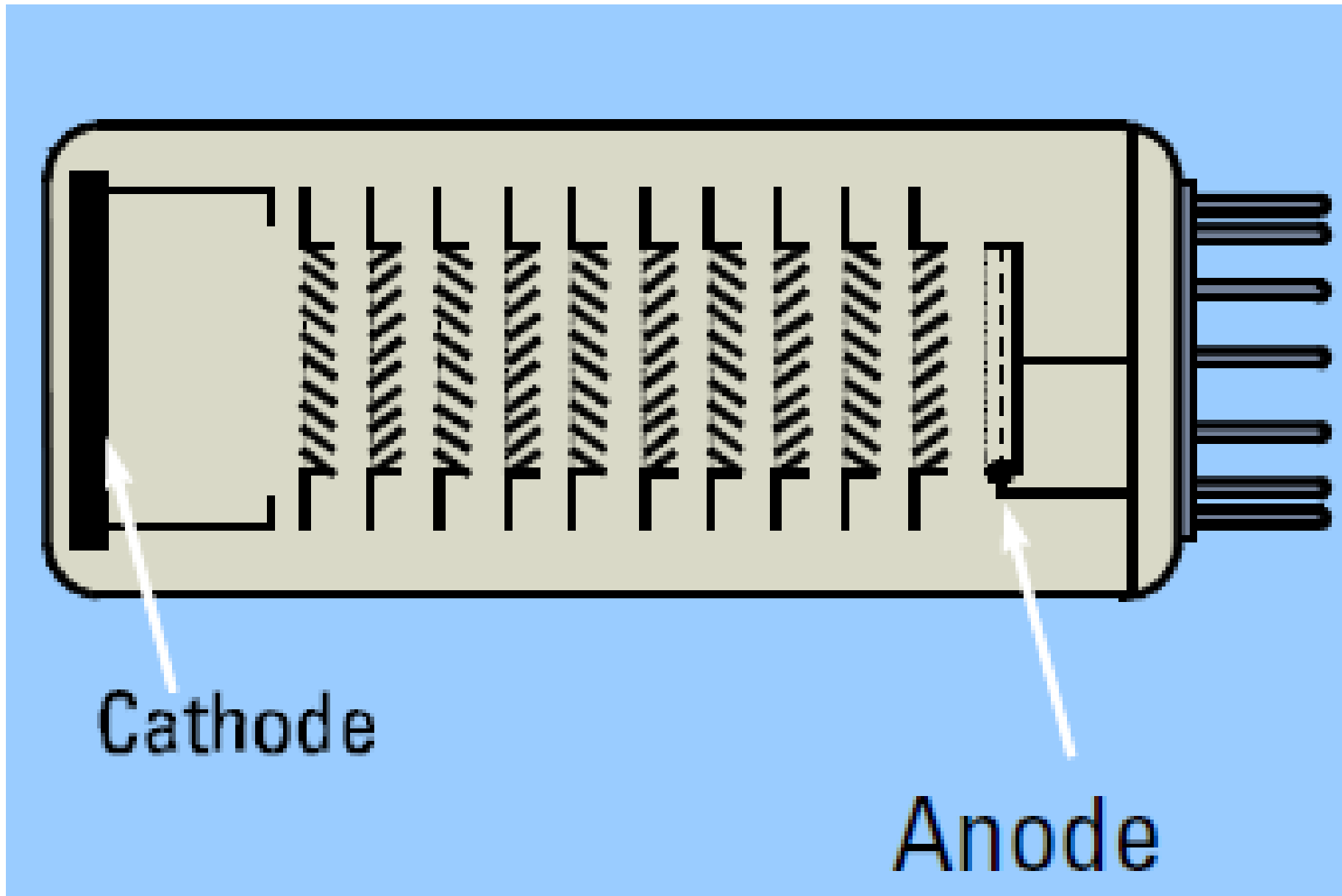


3 ...causing an electric current to flow and giving a reading on the meter.

C) Photomultiplier tube (PMT)

- The most common detector for UV/Visible spectrophotometer is the PMT.
- When light of a particular intensity strike it, current, or electrical signal, of proportional magnitude is generated, amplified, and sent on to the read out.
- It consists of a photo cathode, an anode, and a series of **dynode for multiplying the signal.**

- Light beam strike the photo cathode....e- emitted & accelerated to the 1st dynode.....more e- are emitted...these e- pass on to the 2nd , 3rd etc dynode even more e- are emitted... e- reach the anode, the electrical signal sufficiently multiplied....sent to readout.
- Irradiation of cathode produces electrons, series of anodes (dynodes) increases gain to 10^5 - 10^7 electrons per photon.



Readout

- Where the output is read, converts the electrical signal from the photodetector to a readable form of data.
- Many forms of readout can be used:
 - Computer display
 - Digital or analog readout
 - Strip chart recorders
 - Integrators

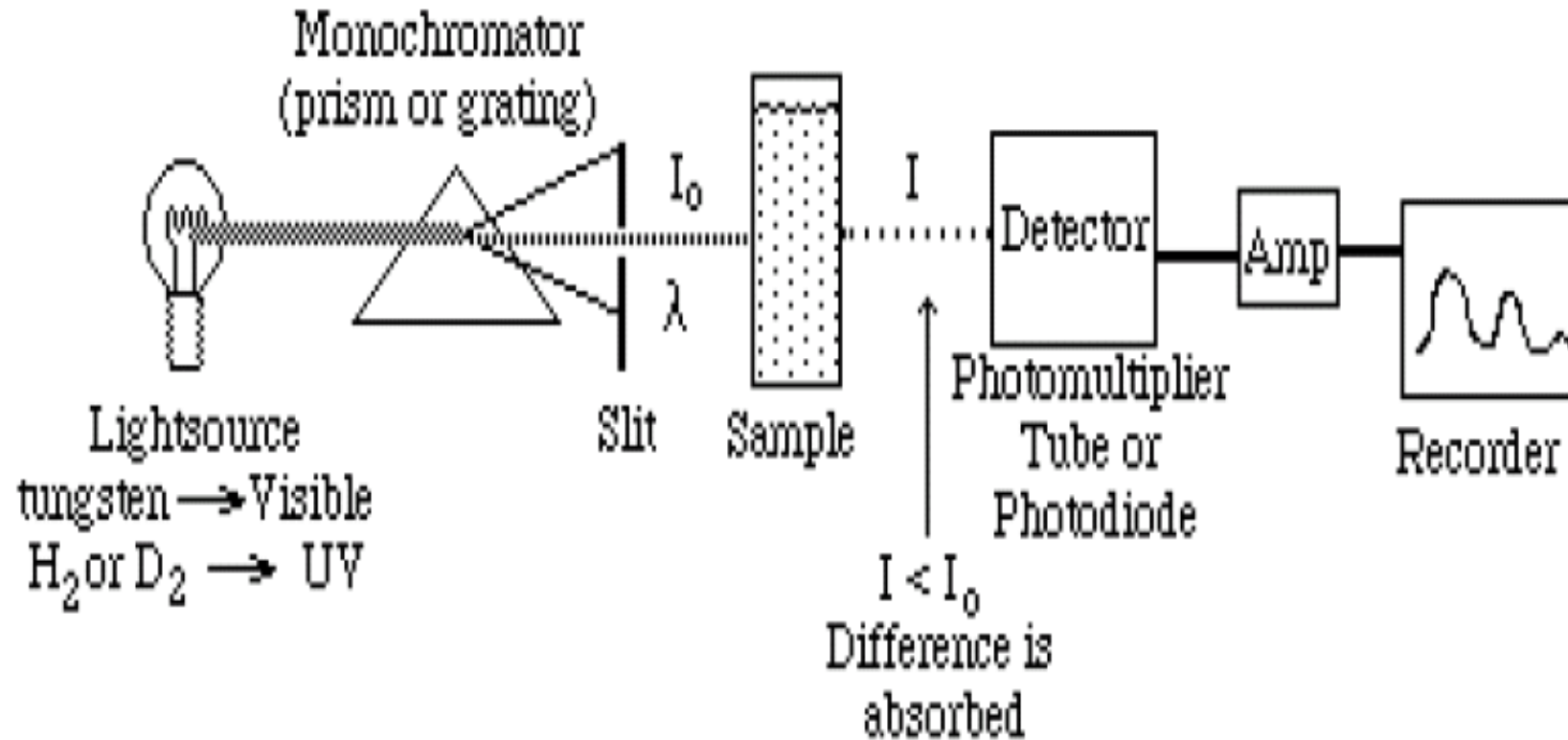
Type of instrument

- ☞ Instrument for measuring the absorption of light may be of single beam or double beam type.

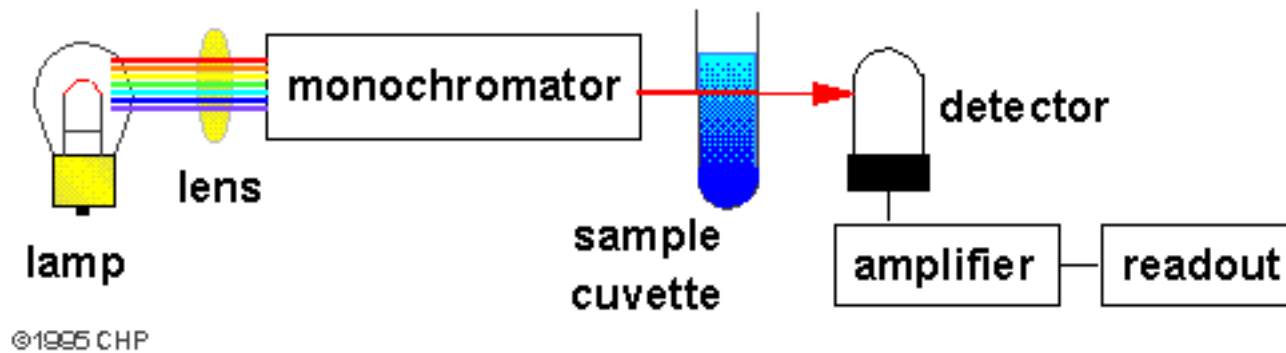
Single-beam instrument

- ☞ Relatively cheap, portable, & ideally suited to quantitative analysis.
- ☞ To record an accurate value of the absorbance it is necessary to zero the instrument on a reference/blank before every measurement.
- ☞ It is essentially a single wavelength measurement of absorbance.

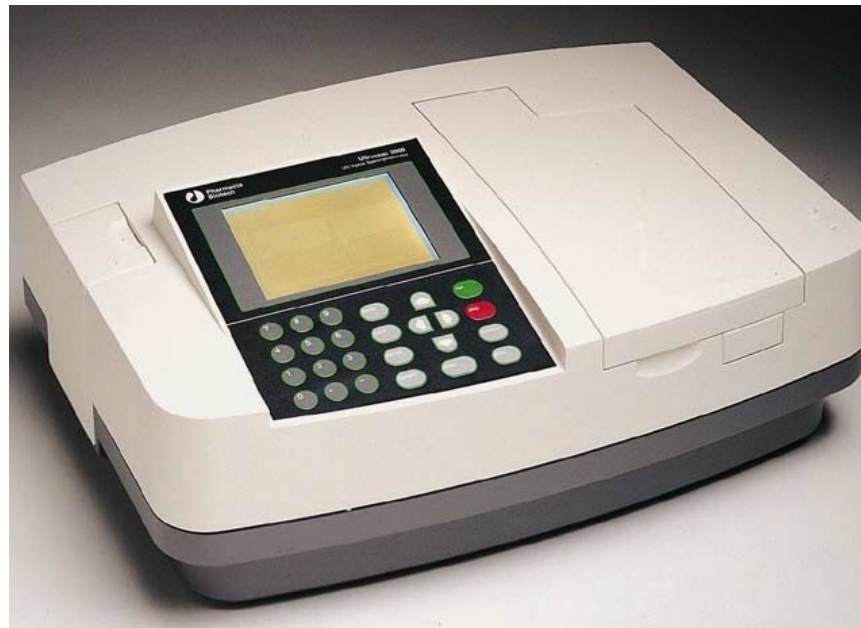
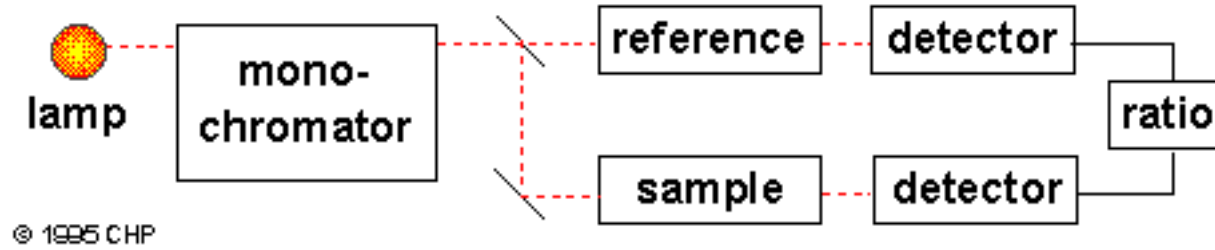
Single Beam Spectrophotometer...



Single Beam Spectrophotometer...



Dual Beam Spectrophotometer...



Double-beam Spectrophotometer ...

- More expensive and complex.
- They are capable of scanning the entire spectrum or make measurement at a single wavelength.
- The signal relayed to the readout is free of effects of intensity fluctuation that cause error.
- The two cuvettes must be matched in terms of path length, refractive & refractive properties.
- Two beams are formed in space by a V-shaped mirror (beam splitter).

Spectrophotometric Applications

- It is one of the most useful tools available to the experts for analysis.
- Important advantages of spectrophotometric methods include:
 1. Wide applicability; large number of organic and inorganic species absorb light in the UV-Visible ranges.
 2. High sensitivity; analysis for concentrations in the range from 10^{-4} to 10^6 M are ordinary in the Spectrophotometric determinations.
 3. Moderate to high selectivity; Due to selective reactions, selective measurements & different mathematical treatments.
 4. Good accuracy; Relative errors in concentration measurement lie in the range of 0.1 to 2 %.
 5. Ease & convenience; Easily & rapidly performed with modern instruments.

Why do we measure UV/VIS spectra?

Why measure?

1. UV/VIS spectra allow components present in the sample solution to be identified.
 - the position and to some extent, the profile of the absorption peaks allow specific compounds to be identified.
 - organic compounds can be identified by their spectra, or solvent purity can be easily checked by UV/VIS spectroscopy.
2. Absorption peaks can be used to quantify the investigated sample.
 - the sample concentration can be calculated from the absorbance value of the peak.

Why measure?

3. Based on the relationship between absorbance and sample concentration, UV/VIS spectroscopy is applied as a quantitative analytical technique in market segments

➤ Water Testing, Food and Beverages, Pharmaceutical, Chemical and Biotech Industry

4. The position of the peaks in the spectrum reveals information about the molecular structure of the sample.

➤ specific functional groups of a molecular structure, such as C=O or C=C, absorb at specific characteristic wavelengths.

Why measure?

- 5. The spectrum may reveal specific physical properties of the sample molecules.
 - ✓ calculate the extinction coefficient of the sample
 - ✓ calculate the melting point of proteins and nucleic acids by measuring the UV/VIS spectra at different temperatures
 - ✓ determine the rate of a reaction by monitoring the absorption spectra as a function of time (also known as kinetic measurements).

Why measure?

6. position and profile of the peaks in the spectrum can give information about the microscopic environment of the sample molecules.

- the presence of impurities or other solvents in the sample solution has an effect on position and of the profile of the peaks.
- the peaks may be broader or have shifted due to impurities.

Applications of Ultraviolet-Visible Spectrophotometry

- UV-visible spectroscopy is a technique that readily allows one to determine the **concentrations of substances**
- Therefore, enables scientists to study
 - The **rates of reactions**, and
 - Determine rate equations for reactions from which a **mechanism** can be proposed.
- UV spectroscopy is used extensively in teaching, research and analytical laboratories for the **quantitative analysis** of all molecules that absorb ultraviolet and visible EMR

Applications of UV-Vis Spectrophotometry...

- **Qualitative analysis**
 - Structure elucidation of organic compounds
 - Detection of impurities
- **Quantitative analysis**
 - Determination of concentration of the compound
 - Chemical kinetics: Study of chemical reactions
 - Dissociation constants of acids and bases
 - In the quantification and thermal denaturation of DNA
- **Study of charge transfer complexes**
- **Other uses**

Qualitative Analysis

Qualitative Applications

- In terms of qualitative analysis of the analyte, the UV-VIS spectrometry is **of a secondary importance** for the *identification and the determination of structural details*.
- The information obtained from it needs to be *supplemented by that from IR, NMR and mass spectrometry*.
- ❖ *it can still provide information* about the presence or absence and the nature of the chromophore in the molecule.
- Features of absorption spectra such as the **molar absorptivity**, **spectral position**, and **shape** and **breadth** of the absorption band are related to molecular structure and environment.

1. Identification of chromophores

- The presence of an absorbance band at a particular wavelength often is a good indicator of the presence of a chromophore.
- Useful information about substance can be obtained via examination of its λ_{max} and ϵ_{max} , which could be correlated with the structural features.

Typical Absorption on Some Common Chromophores (Kristianingrum, 2013).

Chromophores / Compounds	λ_{max} (nm)	ϵ_{max}	Transition
Alkene	177	13000	$\pi \rightarrow \pi^*$
Alkyne	178-225	10000-160	$\pi \rightarrow \pi^*$
Carbonyl	186-280	1000	$n \rightarrow \sigma^*$
Carboxyl	204	41	$n \rightarrow \pi^*$
Amido	214	60	$n \rightarrow \pi^*$
Azo	339	5	$n \rightarrow \pi^*$
Nitro	280	22	$n \rightarrow \pi^*$
Nitroso	300-665	100-20	$n \rightarrow \pi^*$
Nitrate	270	12	$n \rightarrow \pi^*$
Olefin	184	12	Delocalization n^*
Triolefin	250	10000	Delocalization n^*
Diolefin	217	-	Delocalization n^*
Ketones	282	21000	$n \rightarrow \pi^*$
H ₂ O	167	24	$n \rightarrow \sigma^*$
Methanol	184	1480	$n \rightarrow \sigma^*$
Methyl Chloride	173	15	$n \rightarrow \sigma^*$
Dimethyleter	184	200	$n \rightarrow \sigma^*$
Methylamine	215	2500	$n \rightarrow \sigma^*$
Benzene	204	900	$\pi \rightarrow \pi^*$
Toluene	207	7000	$\pi \rightarrow \pi^*$
Phenol	211	6200	$\pi \rightarrow \pi^*$

Qualitative...

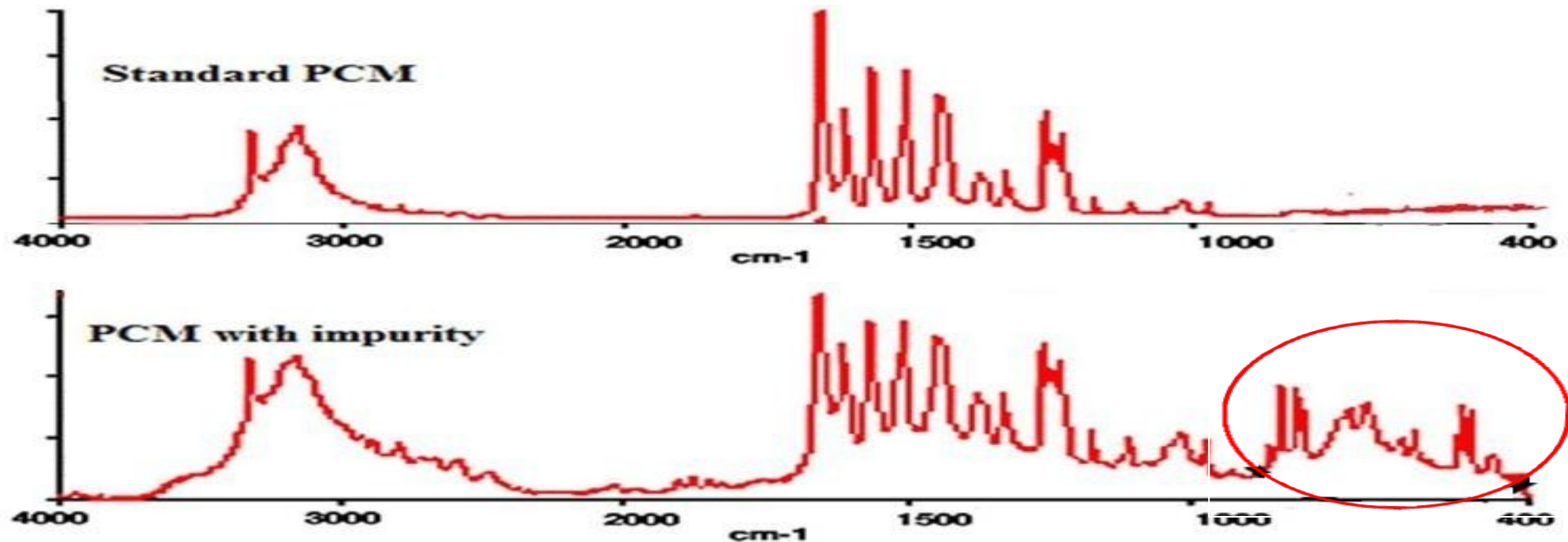
Absorption characteristics of some common organic chromophores:

Chromophore	Example	λ_{max} (nm)	ϵ_{max}
Alkene	R.CH=CH_2	177	13,000
Conjugated alkene	$\text{CH}_2=\text{CH.CH=CH}_2$	214	21,000
Carbonyl	$\text{CH}_3.\text{CO.CH}_3$	186	1000
Carboxyl	$\text{CH}_3.\text{COOH}$	204	41
Azo	$\text{R.CH}_2.\text{N=N.CH}_3$	339	5
Aromatic	Benzene	204 255	7,900 200

Applications of UV-Vis Spectrophotometry...

Detection of Impurities

- UV absorption spectroscopy is one of the best methods for determination of impurities in organic molecules
- Additional peaks can be observed due to impurities in the sample and can be compared with reference standard



BY YOHANNES G

2-Confirmation of identity

- ✓ The spectrum is a physical constant, which along with melting & boiling points, refractive index and other properties may be used for characterization of compounds.
- ✓ Although UV-visible spectra do not enable absolute identification of an unknown, they frequently are used to confirm the identity of a substance:

2. Confirmation of identity...

❖ **Through comparison of the measured spectrum with a reference spectrum.**

- a) An absorption band at 254 nm with characteristic vibrational fine structures may be an evidence for existence of aromatic structure.
- b) Three characteristic bands at 278, 361 & 550 nm with absorbance ratio of 2:3:1 is very characteristic for cyanocobalamin.

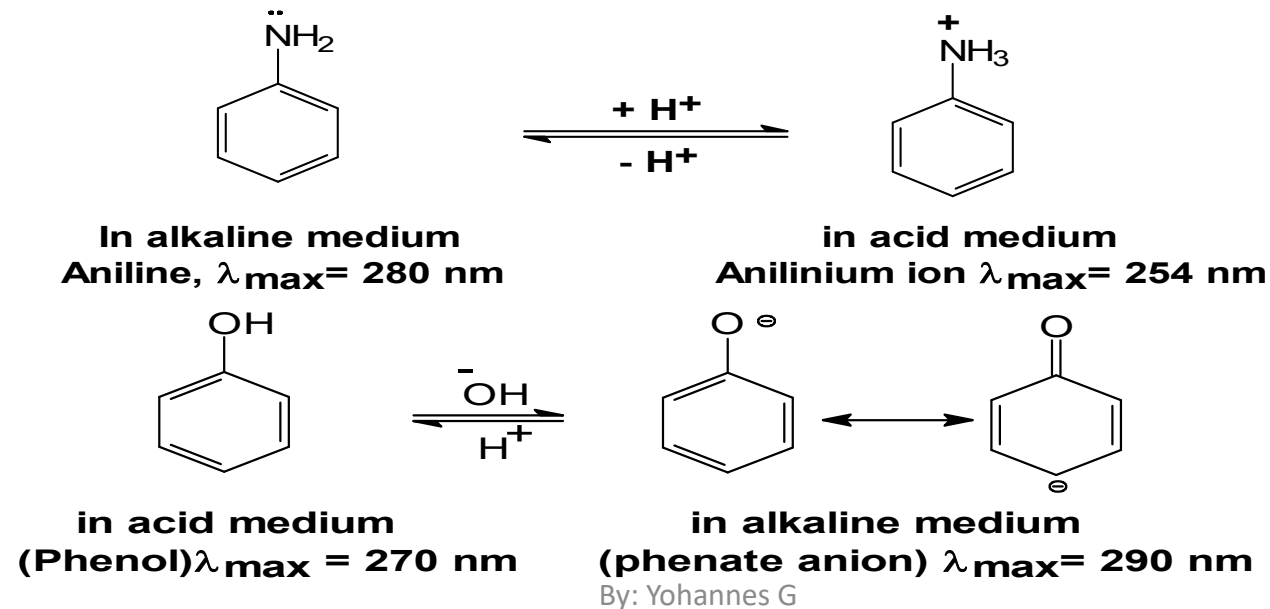
2. Confirmation of identity...

Identification by using Absorbance ratio

- Absorbance ratio of a given drug at two different wavelength is constant, provided that
 - Beer's Law is obeyed at the selected wavelengths
 - The same concentration of the sample is used for both wavelengths

3.Detection of some functional groups

- a) An absorption band at about 280-290 nm, which is displaced toward shorter wavelength with increasing solvent polarity strongly, indicates the presence of aromatic carbonyl group.
- b) Confirmation of presence of aromatic amine or phenolic structure may be obtained by testing the pH effect on their spectra.



4. Approximate determination of the number of double bonds:

➤ By using simplified Kuhn and Hausser rule

5. Identification of the position and/or conformation of certain functional groups:



- α -Alkyl cause red shift about 10 nm & α -OH about 35 nm
- β -Alkyl cause red shift about 12 nm & β -OH about 30 nm
- γ -Alkyl cause red shift about 18 nm & γ -OH about 50 nm

Quantitative Analysis

Quantitative...

- UV-visible spectrophotometry is perhaps the most widely used spectrophotometric technique for the quantitative analysis of chemical substances as pure materials and as components of dosage forms.
- It has found increasing usefulness as a means of assaying pharmaceutical substances described in the pharmacopeias.
- The use of UV-visible spectrophotometry in quantitative analysis is centered around the **Beer-Lambert law**, which relates the absorbance of an analyte to its concentration.

Quantitative...

- ❑ **Beer's law**, tells us quantitatively how the amount of attenuation depends on the concentration of the absorbing molecules and the path length over which absorption occurs.
- As light traverses a medium containing an absorbing analyte, the intensity decreases as the analyte becomes excited.

For an analyte solution of a given concentration, the longer the length of the medium through which the light passes (path length of light).

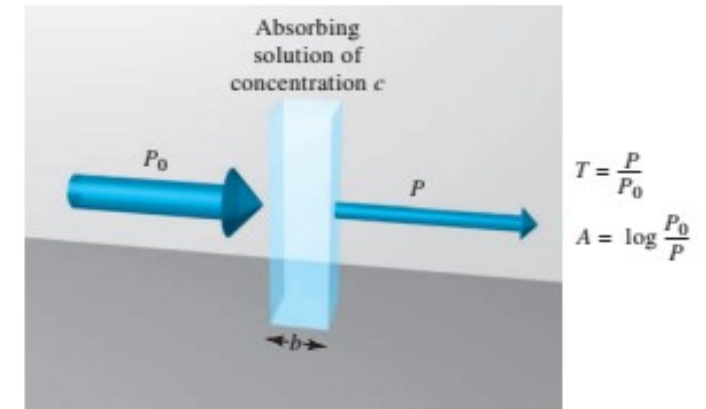
- ✓ The more absorbers are in the path, and the greater the attenuation.
- ✓ For a given path length of light, the higher the concentration of absorbers, the stronger the attenuation.

Quantitative...

- Beer's law depicts the attenuation of a parallel beam of **monochromatic radiation** as it passes through an absorbing solution of thickness b , cm and concentration c moles per liter.
- Because of interactions between the photons and absorbing particles, the radiant power of the beam decreases from P_0 to P .
- The **transmittance** T of the solution is the fraction of incident radiation transmitted by the solution.
- Transmittance is often expressed as a percentage and called the **percent transmittance**.
- Multiplying the transmittance by 100 gives the percent transmittance (%T), which varies between 100% (no absorption) and 0% (complete absorption).
- $T = P/P_0$

Quantitative...

Absorbance



- The **absorbance**, A , of a solution is related to the transmittance in a logarithmic manner.
- As the absorbance of a solution increases, the transmittance decreases.

$$A = -\log T = -\log \frac{P}{P_0} = \log \frac{P_0}{P}$$

Quantitative ...

Beer's-Lambert's Law

➤ This is a combination of both laws ; $\text{Log } I_0/I_t = A = abC$

where

a is a constant called absorptivity,

b is the path-length in cm

C is the concentration in grams/Liter.

➤ The value of a will clearly depend upon the method of expression of the conc.

➤ If C is expressed in moles/liter, and b in cm then a is given the

symbol ϵ (Epsilon, $\text{l.mol}^{-1}\text{cm}^{-1}$) and is called the molar

absorption coefficient or molar absorptivity.

$A = \epsilon b C$. When ϵ is measured at λ_{max} it is called ϵ_{max} .

Beer's law states that, using a monochromatic wavelength, absorbance is directly proportional to concentration.

Quantitative...

- Absorbance is the more common unit for expressing the attenuation of radiation because it is a linear function of the analyte's concentration.
- Besides absorption by the analyte, several additional phenomena contribute to the net attenuation of radiation, including reflection and absorption by the sample container, absorption by components of the sample matrix other than the analyte, and the scattering of radiation.
- ✓ To compensate for this loss of the electromagnetic radiation's power, we use a method blank.

Quantitative ...

- The **extinction coefficient** (ϵ) is characteristic of a given substance under a precisely defined set of conditions, such as **wavelength**, **solvent**, and **temperature**.
- In practice, the measured extinction coefficient also depends partially on the characteristics of the instrument used.
- For these reasons, predetermined values for the extinction coefficient usually are not used for quantitative analysis.
- Instead, a calibration or working curve for the substance to be analyzed is constructed using one or more standard solutions with known concentrations of the analyte.

Quantitative ...

Assumptions of the absorption law:

- The incident beam is monochromatic
- The absorbers absorb independently of each other.
- Incident radiation consists of parallel rays perpendicular to the surface of the absorbing medium.
- Path length traversed is uniform over the cross section of the beam.
- Absorbing medium is homogenous and does not scatter the radiation.

Quantitative ...

❖ **Deviations from the direct proportionality between the measured absorbance and concentration when path length is constant may be encountered.**

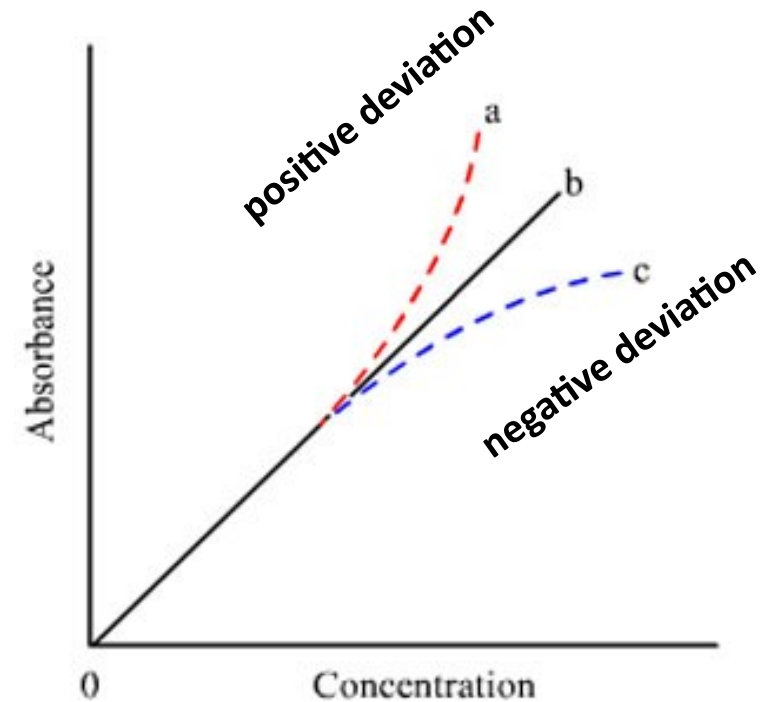
- ✦ A plot of absorbance versus concentration is expected to be a straight line passing through origin.
- ✦ The law does not hold true for all species under every condition.
- ✦ Many a times instead of a straight line, a curvature in the plot may be observed as shown in figure below.

Quantitative ...

- Some of the factors responsible for the deviation from Beer's law are as follows.

Presence of Electrolytes

- The presence of small amounts of colorless electrolytes which do not react chemically with the colored components does not affect the light absorption as a rule.
- However, large amounts of electrolytes may affect the absorption spectrum qualitatively as well as quantitatively.

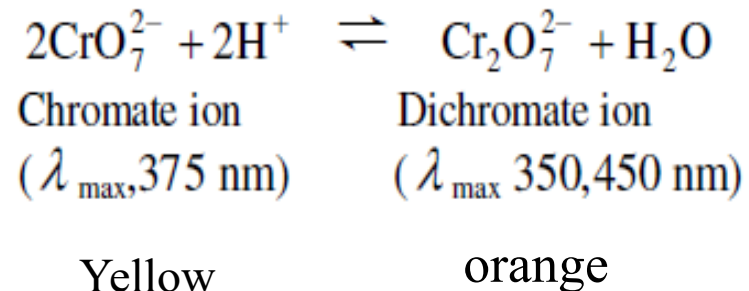


Quantitative ...

- This is due to the **physical interaction** between the ions of the electrolyte and the colored ions or molecules.
- This interaction results in a deformation of the colored ions or molecules
- Thereby causing a change in its light absorption property.

H⁺ Concentration

- There are a number of substances whose ionic state in solution is greatly influenced by the presence of H⁺.

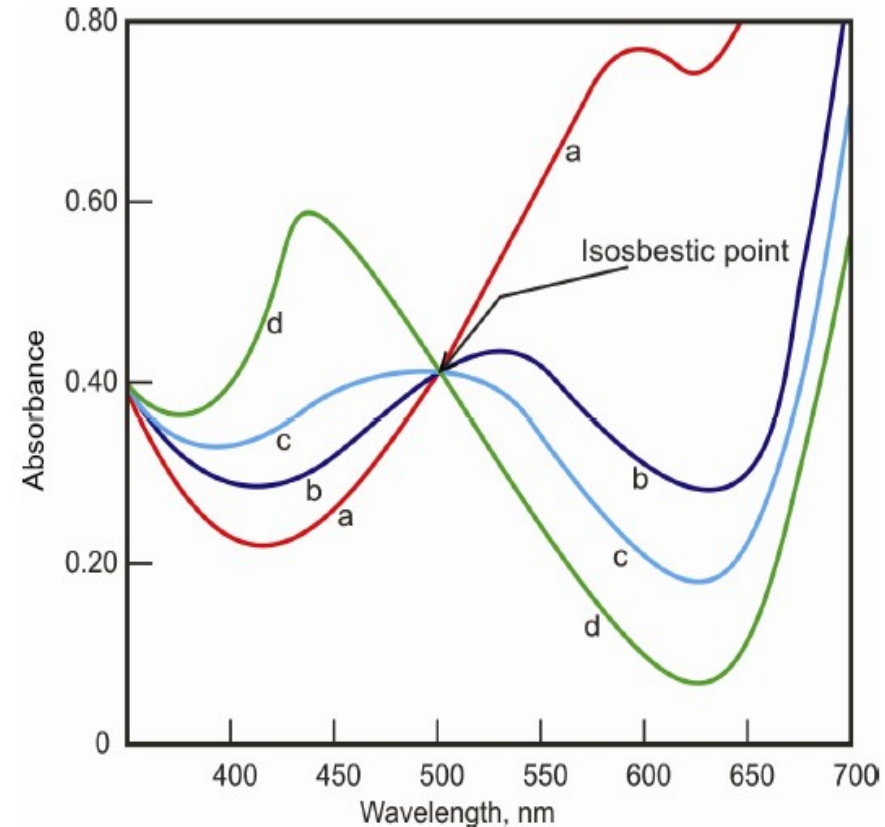


Quantitative ...

- In some cases, two absorbing species are in equilibrium and have a common value of absorptivity at a certain wavelength.
- For example, in case of bromothymol blue the absorption spectra at different pH values are different.
- However, at wavelength of 501 nm, we see that all species have same molar absorptivity .

Quantitative ...

- Such a wavelength is known as **isosbestic point**.
- At this wavelength the Beer's law holds, though the measurements have low sensitivity.
- However, such wavelengths should be avoided for the quantitative work.

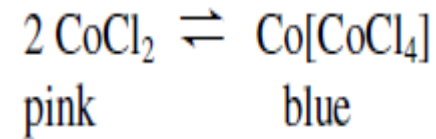


Quantitative ...

Complexation, Association or Dissociation

- Some salts have a tendency to form complexes whose colours are different from those of the simple compounds.

For example, the colour of cobalt chloride changes from pink to blue due to the following complex formation.



- The degree of complex formation increases with increase in conc., therefore, Beer's law does not hold at high concentrations. .

Quantitative ...

Non-monochromatic Nature of the Radiation

- In order for the Beer's law to hold, it is necessary that monochromatic light is used.
- The wider the bandwidth of radiation passed by the filter or other dispersing devices, the greater will be the apparent deviation of a system from adherence to Beer's law.

Concentration of the Analyte

- As per Beer and Lambert's law, the plot of A versus the conc. should be a straight line when ϵ and b are constant.
- Therefore, at higher conc (**1 mmol/dm³**) there may be deviation from the law.

Quantitative ...

- This is because interactions b/n absorbing species at higher conc affect ability to absorb radiation due to:
 - ✓ crowding,
 - ✓ molecular interaction and association
 - ✓ charge distribution

Temperature

- The T is not considered as an important factor since ordinarily the measurements are made at a constant T.
- However, changes in T sometimes may shift ionic equilibrium and the absorptivity.
- For example, the colour of acidic FeCl_3 solution changes from yellow to reddish brown on heating due to change in λ_{max} and absorptivity.

Quantitative ...

(1) Real deviation:

- Valid only for dilute ($< 10^{-3}$ M) solutions. Why?
 - ✓ Because interactions between absorbing species at higher concentrations affect ability to absorb radiation due to crowding, molecular interaction and association as well as charge distribution
 - ✓ Because also ϵ depends on **refractive index** which depends on concentration of the solution (mostly at high concentration levels)

(2) Instrumental deviation(errors):

- a) **Irregular deviation** due to: Unmatched cells, unclean handling and unclean optics.
- b) **Regular deviation:** due to:
 - 1- Errors in wavelength scale and Slit width control.

Quantitative ...

2- Stray light

- Stray light is light which falls on detector within a UV instruments without passing through the sample
- It can arise either from light scattering within the instrument or by entry of light into the instrument from outside
- It gives a false low absorbance reading for the sample since it appears as though the sample is absorbing less light than it actually is.
- This is most serious where the sample has a high absorbance e.g. at an absorbance of 2 the sample is absorbing most of the light passing through it and thus it would only require very low intensity stray light to lower the reading substantially.

Quantitative ...

c) Other errors:

- ✓ Non-linear response of photo cells
- ✓ Radio and TV interferences
- ✓ Un stabilized power supply

Assay of single components

- The assay of an absorbing substance may be quickly carried out by preparing a solution in a transparent solvent and measuring its absorbance at a suitable wavelength.
 - Wavelength of maximum absorbance
- The concentration of the absorbing species is then calculated from the measured absorbance using the following procedures.
 - i. Use of standard absorptivity value-** used for substances whose reference standards are expensive or difficult to obtain.

- **Example 1:** Calculate the concentration of methyl testosterone in an ethanolic solution of which the absorbance in a 1 cm cell at its λ_{max} , 241 nm, was found to be 0.890. The specific absorptivity in the B.P. is 540 at 241 nm. (0.00165 g/100ml)

Example 2: calculate the concentration in ug/ml of a solution of tryptophan (M.wt = 204.2) in 0.1 M HCl, giving an absorbance at its λ_{max} , 277 nm, of 0.613 in 4 cm cell. (molar absorptivity is 5432). (5.76 ug/ml)

- **Example-3:** A 5.00×10^{-4} M solution of an analyte is placed in a sample cell that has a path length of 1.00 cm. When measured at a wavelength of 490 nm, the absorbance of the solution is found to be 0.338. What is the analyte's molar absorptivity at this wavelength? **Ans(Molar A. = $676 \text{ cm}^{-1} \text{ M}^{-1}$)**

Example -4: A sample has a percent transmittance of 50.0%. What is its absorbance? **Ans (A= 0.301)**

Example-5: The molar absorptivity of a substance is $2.0 \times 10^4 \text{ cm}^{-1} \text{ mol}^{-1} \text{ L}$. Calculate the transmittance through a cuvette of path length 5.0 cm containing $2.0 \times 10^{-6} \text{ mol L}^{-1}$ solution of the substance.

Ans (T= 0.63)

ii. Use of calibration graph

$$Y = ax + b$$

Example: the absorbance values at 250 nm of 5 standard solutions, and sample solution of a drug are given below:

Conc. (ug/ml)	A 250 nm
10	0.168
20	0.329
30	0.508
40	0.660
50	0.846

Sample 0.661

Calculate the concentration of the sample.

($Y = 0.01679X - 0.0008$, $C = 36.5$ ug/ml)

iii. Single point standardization

- this method involves the measurement of the absorbance of a sample solution and of a standard solution of the reference substance.
- The standard and sample solution are prepared in similar manner.
- The concentration of the substance in the sample is calculated from the proportional relationship that exists b/n absorbance and concentration.

$$\mathbf{C\ sample = A\ sample \times C\ standard / A\ standard}$$

iii. Single point ...

- It is the procedure adopted in many spectrophotometric assays of the BP and for the majority of spectrophotometric assays of USP.
- Occasionally, a linear but non proportional relationship b/n concentration and absorbance occurs, which is indicated by a significant negative or positive intercept in a beer's law plot.

iv. Double point standardization

- a two –point bracketing standardization is therefore required to determine the concentration of the sample solution.
- The concentration of one of the standard solution is greater than that of the sample while the other is with a` lower concentration than the sample.
- The concentration of the test sample is given by:

$$C_{\text{test}} = \frac{(A_{\text{test}} - A_{\text{std1}})(C_{\text{std1}} - C_{\text{std2}}) + C_{\text{std1}}(A_{\text{std1}} - A_{\text{std2}})}{A_{\text{std1}} - A_{\text{std2}}}$$

- Where, **Std1** is the more concentrated standard and **Std2** is less concentrated standard

Multicomponent Analysis

- The spectroscopic analysis of drugs rarely involves measurement of absorbance of samples containing only one absorbing component.
- The pharmaceutical analyst frequently encounters the situation where the concentration of one or more substance is required in samples known to contain other absorbing substances which potentially interfere in the assay.

Multicomponent ...

- The basis of all the spectrophotometric techniques for multicomponent samples is the property that:
 - a) The absorbance of a solution is the sum of absorbances of the individual components or
 - b) The measured absorbance is the difference b/n the total absorbance of the solution in the sample cell and that of the solution in the reference cell.

Multicomponent ...

- **If the identity, concentration and absorptivity of the absorbing interferents are known:**

➤ it is possible to calculate their contribution to the total absorbance of a mixture.

Example: The max of ephedrine HCl and Chlorocresol are 257 nm and 279 nm respectively and the specific absorptivity values in 0.1M HCl solution are:

- Ephedrine HCl at 257 nm = 9.0 Chlorocresol at 257 nm = 20
- Ephedrine HCl at 279 nm = 0 Chlorocresol at 279 nm = 105

Calculate the concentration of ephedrine HCl and Chlorocresol in a batch of Ephedrine HCl injection, diluted 1 to 25 with water, giving the following absorbance values in 1 cm cells.

ü (A₂₇₉ (total) = 0.424, A₂₅₇ (total) = 0.97

ü **Ans. C. Ephedrine HCl in the injection = 2.475 gm/100ml**

C. Chlorocresol = 0. 1010 gm/100ml

Multicomponent...

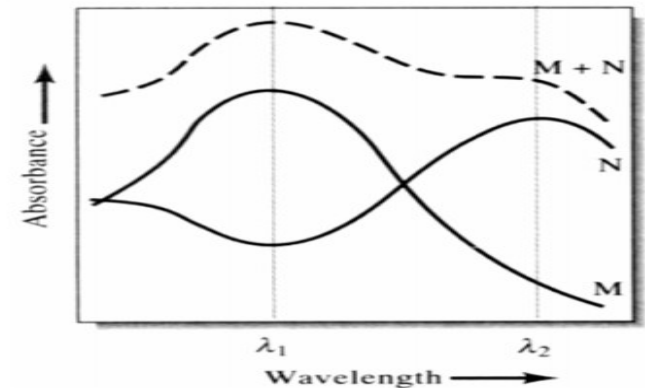
- The absorbance of the mixture at each wavelength is the sum of the absorbance of each component at that wavelength
- Thus for two components x and y , the equations are:

$$A'_{(x+y)} = A'_x + A'_y = e'_x b c_x + e'_y b c_y$$

$$A''_{(x+y)} = A''_x + A''_y = e''_x b c_x + e''_y b c_y$$

where

- ✓ A' is absorbance at wavelength $'$
- ✓ A'' is absorbance at wavelength $''$
- ✓ e' and e'' is molar absorptivity at wavelength $'$ and $''$
- ✓ c is concentration, and b is path length.



Difference Spectrophotometer

- the selectivity and accuracy of Spectrophotometric analysis of samples containing absorbing interferences may be markedly improved by the technique of **difference spectrophotometer**.

Principle: a component in a mixture is analyzed by carrying out a reaction which is selective for the analyte.

Difference Spectrophotometer...

- bringing about a shift in wavelength through adjustment of pH of the solution in which the analyte is dissolved or a chemical reaction such as oxidation or reduction.
- The measured value is the difference absorbance (ΔA) b/n two equimolar solutions of the analyte in different chemical forms which exhibit different spectral characteristics.

Difference Spectrophotometer...

- ❑ The criteria for applying difference spectrophotometry to the assay of a substance in the presence of other absorbing substances are that:
 - Reproducible changes may be induced in the spectrum of the analyte by the addition of one or more reagents
 - The absorbance of the interfering substance is not altered by the reagents.
- ✓ The simplest and most commonly employed technique for altering the spectral characteristics of the analyte is the **adjustment of the pH** by means of aqueous solutions of acid, alkali or buffers.

Difference Spectrophotometer...

- $\Delta A = A_{\text{alk}}(\text{total}) - A_{\text{acid}}(\text{total})$
 - $\Delta A = \Delta \epsilon \cdot b \cdot C$
- If the substance is not affected by pH, it can be quantitatively converted by means of a suitable reagent to a chemical species that has d/t spectral properties to its unreacted parent species.

Derivative spectroscopy

- ❖ The main effect of derivatization is to remove underlying broad absorption bands where there is only a gradual change in slope.
- Derivative spectroscopy uses first or higher derivatives of absorbance with respect to wavelength for qualitative analysis and for quantification.
- If a spectrum is expressed as absorbance, A , as a function of wavelength, the derivative spectra are

Zero order

$$A = f(\lambda)$$

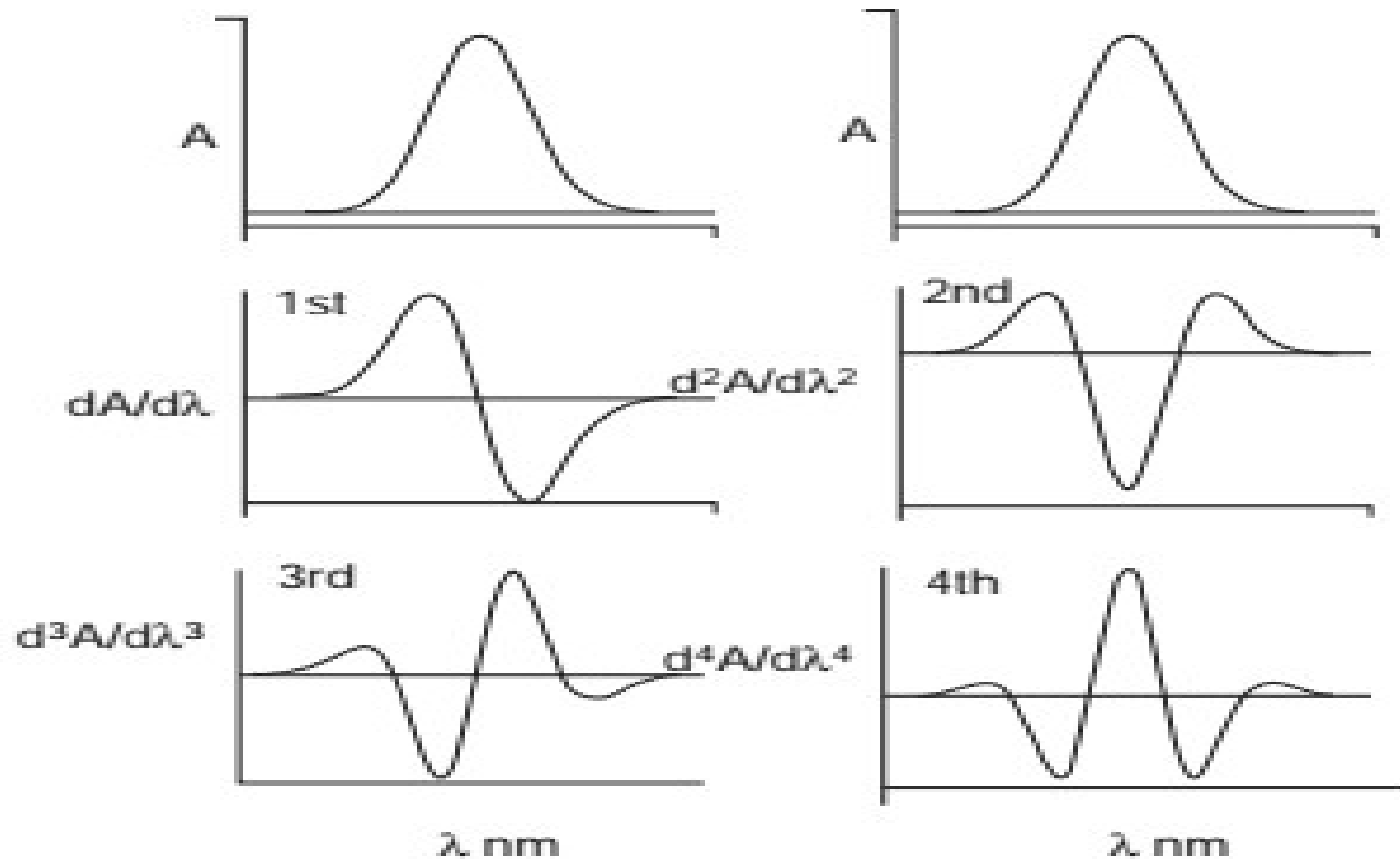
First order

$$\frac{dA}{d\lambda} = f'(\lambda)$$

Second order

$$\frac{d^2A}{d\lambda^2} = f''(\lambda)$$

Derivative spectroscopy...



Derivative spectroscopy...

- A **first-order derivative** is the rate of change of absorbance with respect to wavelength.
- ✓ It passes through zero at the same wavelength as λ_{max} of the absorbance band.
- ✓ This is characteristic of **all odd-order** derivatives.
- The most characteristic feature of a **second-order derivative** is a negative band with minimum at the same wavelength as the maximum on the zero-order band.

Derivative spectroscopy...

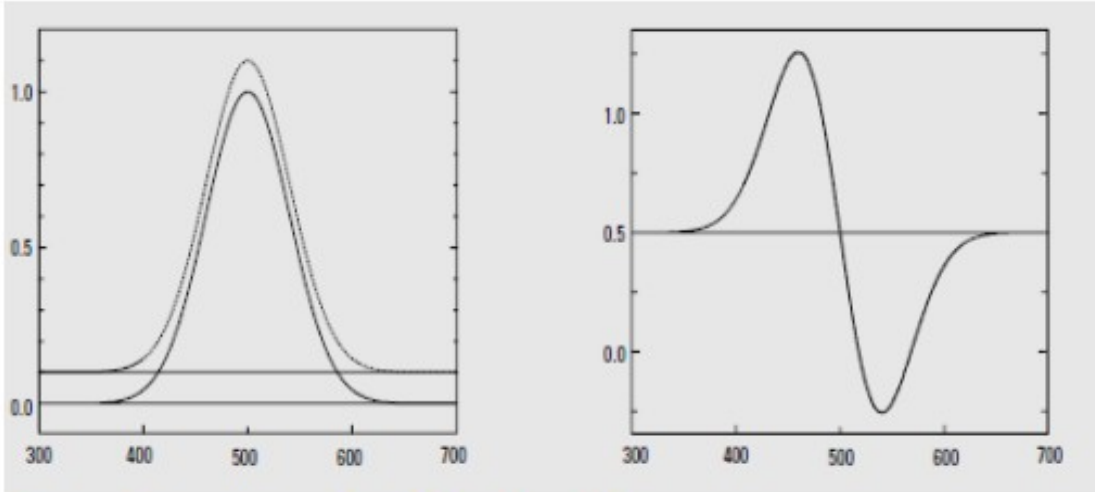
- ✓ A **fourth-order** derivative shows a positive band.
- A strong negative or positive band with minimum or maximum at the same wavelength as λ max of the absorbance band is characteristic of the even-order derivatives.
- ❖ The number of bands observed is equal to the derivative order plus one.

Derivative spectroscopy...

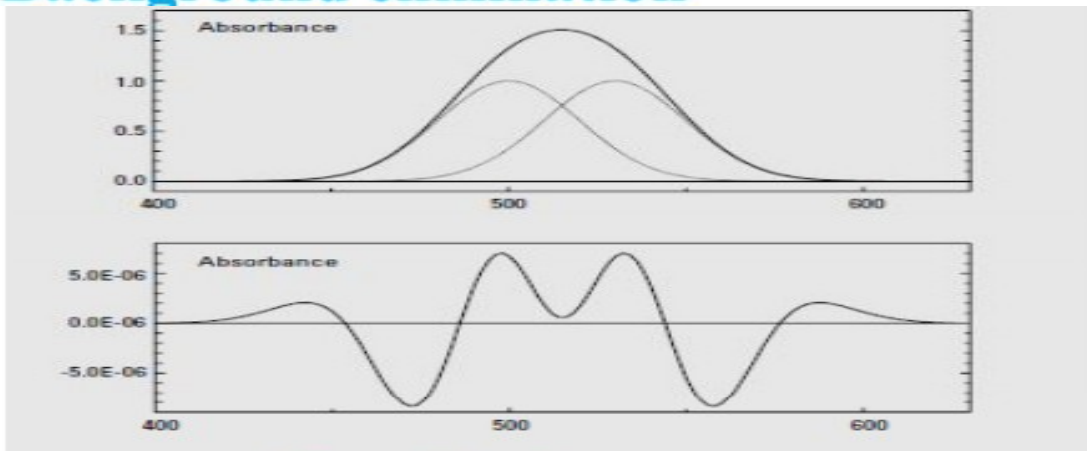
Advantages

- shows better resolution of overlapping bands of the fundamental spectrum and may permit the accurate determination of the λ max of the individual bands.
- It permits discrimination against broad band interferences, arising from turbidity or non-specific matrix absorption.
- ✓ Thus, the information content of a spectrum is presented in a potentially more useful form.
- ✓ A convenient solution to a number of analytical problems,
 - resolution of multicomponent systems, removal of sample turbidity, matrix background and enhancement of spectral details.

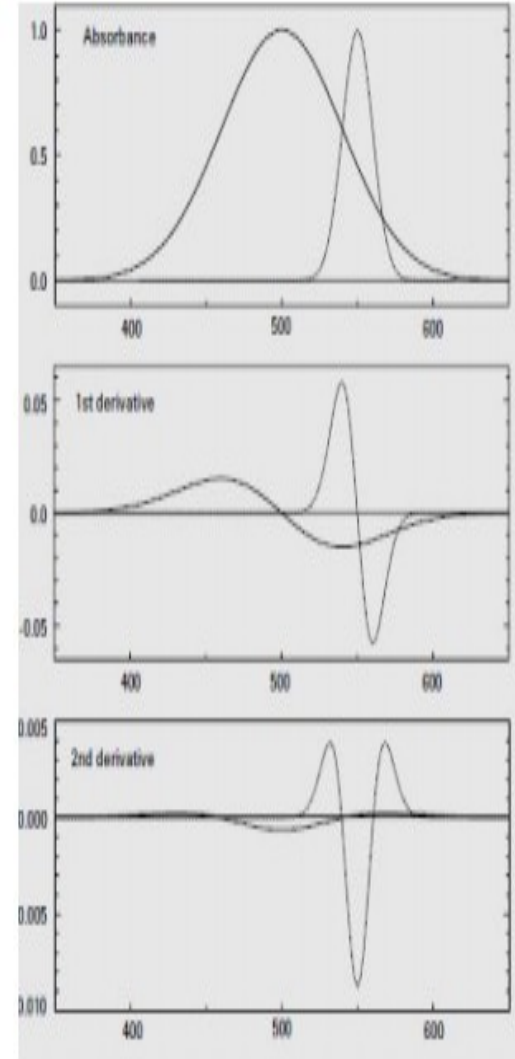
Derivative spectroscopy...



Background elimination



Resolution



Discrimination

Derivative spectroscopy...

- It is possible to measure the absolute value of the derivative at an abscissa value (wavelength) corresponding to a zero crossing of one of the components in the mixture.
- This is termed a **zero-crossing measurement**.
- The zero-crossing derivative spectroscopic mode allows the resolution of binary mixtures of analytes by recording their derivative spectra at wavelengths at which one of the components exhibits no signal.
- Zero-crossing measurements for each component of the mixture are therefore the sole function of the concentration of the others.

Other Applications

A. Monitoring drug degradation kinetics

- Can be simply done when the product has a different absorption spectrum than that of un-degraded drug.
- The rate of disappearance of the spectrum or appearance of other spectrum (as a function of time) may be used to determine rate constant for hydrolysis or degradation.
- Oxidation reactions and any other type of reactions that yield products whose spectra are different from the reactants , may be followed and their rate constant estimated.

Other Applications...

B. Detection in Chromatography

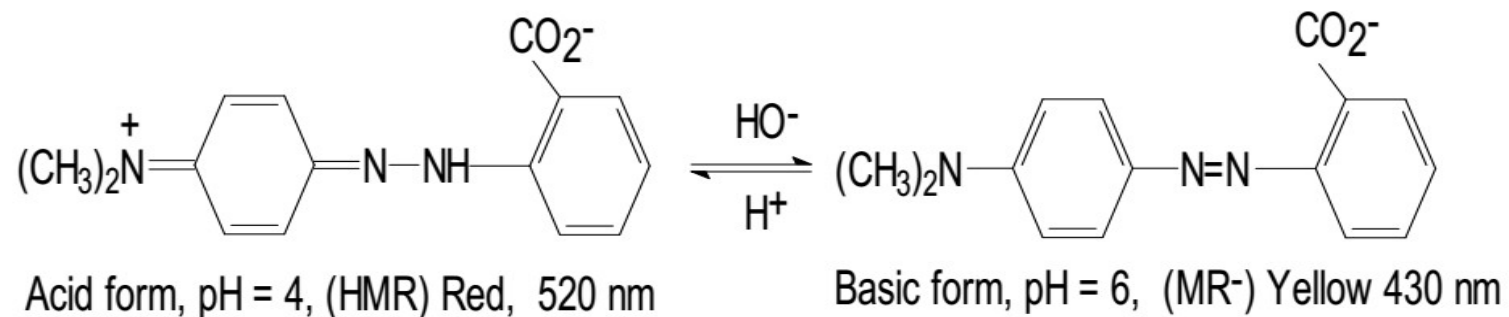
- Mainly used in HPLC and HPTLC.
- They are the most widely used detectors, because:
 - ✓ Most drugs absorb UV-Visible radiation.
 - ✓ More sensitive and more selective than the bulk property detectors (e.g. R.I. detectors).
 - ✓ Some absorbance detectors have one or two fixed wavelengths (280 and/or 254 nm).
 - ✓ More modern HPLC instruments have variable wavelength detectors using the photodiodes

Other Applications...

C. Determination of Equilibrium Constants

- Acid dissociation constants and metal ion-ligand stability constants can be determined spectrophotometrically if the species involved have absorptivity's which differ from one another.

Example : Determination of the pK_a of Methyl red indicator ; Acidic (HMR) and basic (MR⁻) forms of methyl red are shown below



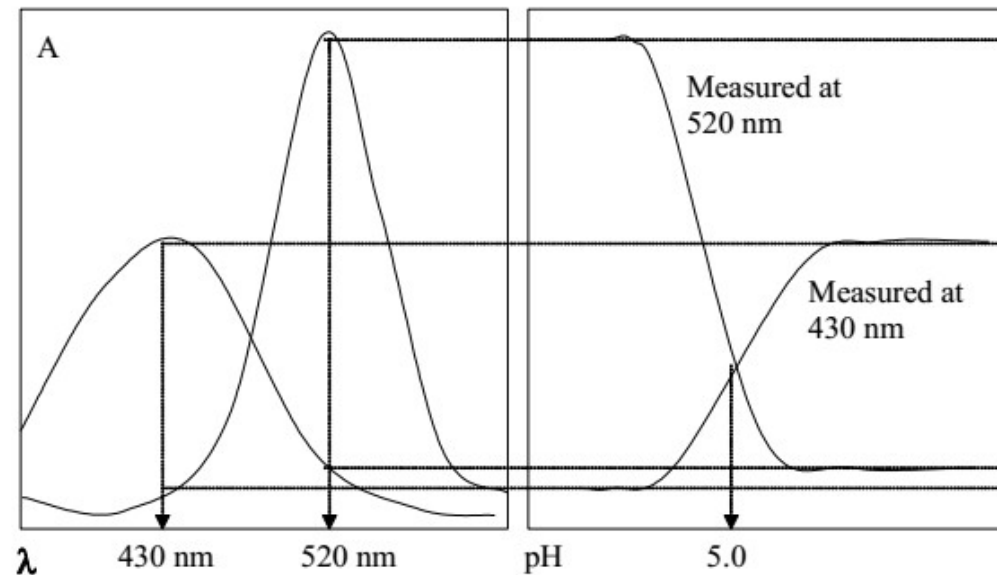
Other Applications...

C. Determination of Equilibrium Constants....

- The pKa of methyl red indicator is given by the equation:

$$\text{pK}_a = \text{pH} - \log [\text{MR}^-] / [\text{HMR}]$$

- ✓ Both HMR and MR⁻ have strong absorption peaks in the visible portion of the spectrum.



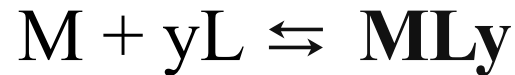
Other Applications...

- The color change interval from pH 4 to pH 6 can be obtained with acetate buffer system.
- At pH = 4, the acid is completely unionized and at pH = 6, the acid is completely ionized.
- At intermediate pH values, the two species are present.
- Plotting absorbance (A) against pH values at 11 and 12 gives two curves.
- The pH at the point of intersection represents the pKa of the indicator.

Other Applications...

D. Determination of complex stoichiometry

- The stoichiometry for a metal–ligand complexation reaction of the following general form.



- Can be determined by one of three methods: the method of continuous variations, the mole-ratio method, and the slope ratio method.

i. Method of continuous variations (CVM)

- ✓ Also called Job's method, is the most popular.
- ✓ In this method a series of solutions is prepared such that the total moles of metal and ligand, n_{tot} , in each solution is the same.
- ✓ Thus, if $(nM)_i$ and $(nL)_i$ are, respectively, the moles of metal and ligand in the i -th solution, then

$$n_{\text{tot}} = (nM)_i + (nL)_i$$

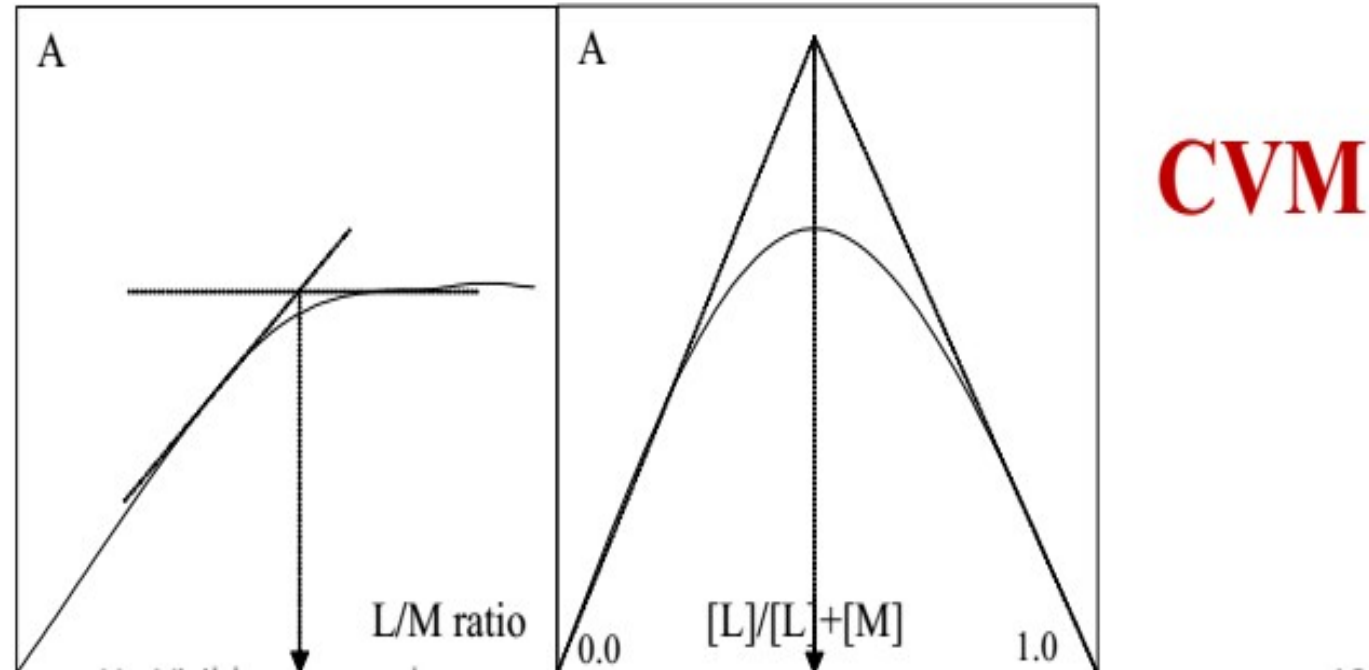
i. Method of continuous variations...

- The relative amount of ligand and metal in each solution is expressed as the mole fraction of ligand, $(X_L)_i$, and the mole fraction of metal, $(X_M)_i$.

$$\checkmark (X_L)_i = \frac{(n_L)_i}{n_{tot}}$$
$$\checkmark (X_M)_i = 1 - (X_L)_i = \frac{(n_M)_i}{n_{tot}}$$

i. Method of continuous variations...

- Absorbance versus the mole fraction of ligand will be plotted.



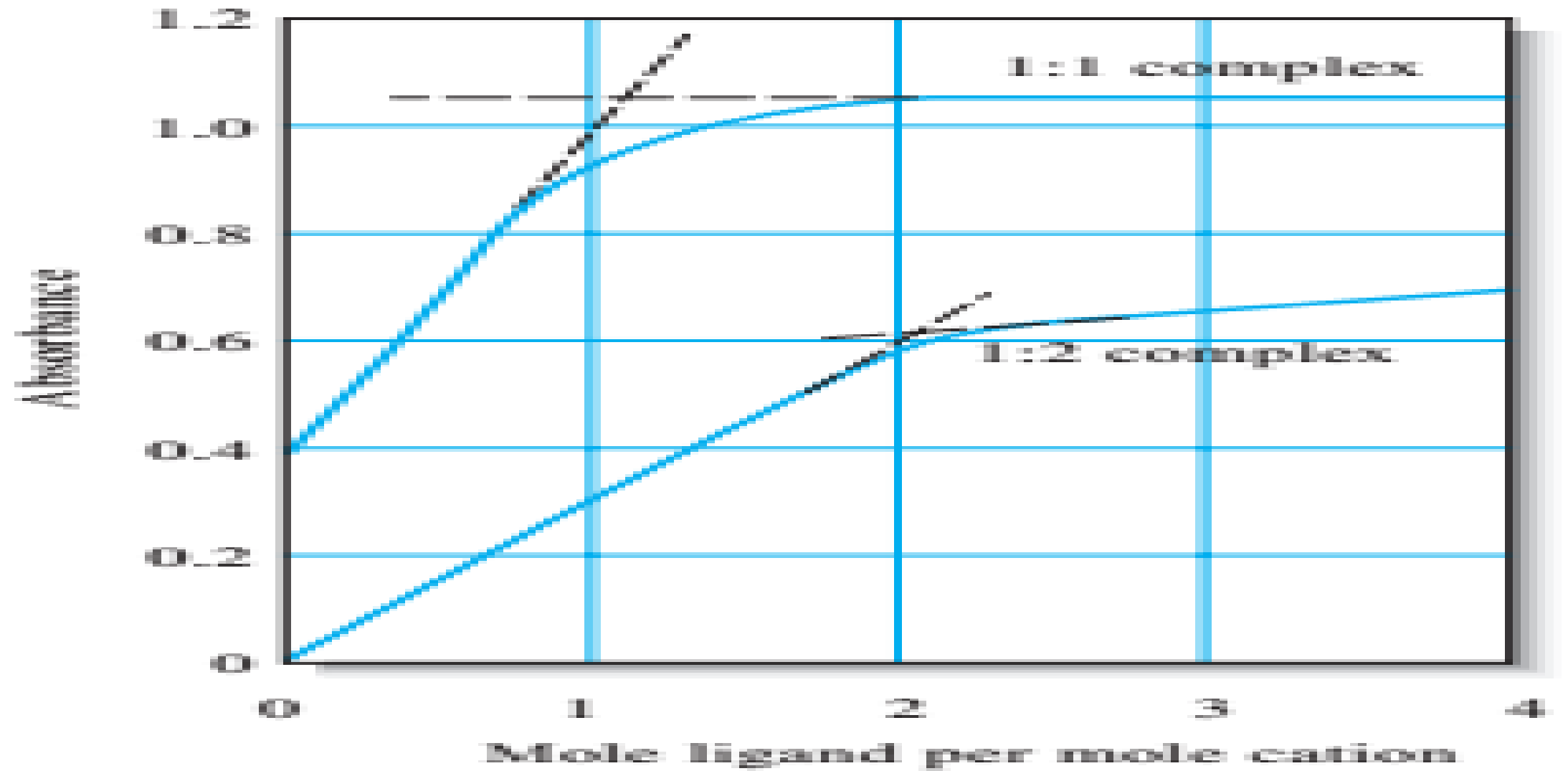
i. Method of continuous variations...

- The intersection of the two lines drawn from both sides occurs when stoichiometric mixing of metal and ligand is reached.
- Mole fraction of ligand at this intersection is used to determine the value of y for the metal–ligand complex, Mly .

- $$Y = \frac{nL}{nM} = \frac{XL}{XM} = \frac{XL}{1-XL}$$

ii. Mole-ratio method

- the moles of one reactant, usually the metal, are held constant, while the moles of the other reactant are varied.
- The absorbance is monitored at a wavelength at which the metal–ligand complex absorbs.
- A plot of absorbance as a function of the ligand-to-metal mole ratio (n_L/n_M) has two linear branches that intersect at a mole ratio corresponding to the formula of the complex.



iii. slope-ratio method

- Two sets of solutions are prepared.
- The first set consists of a constant amount of metal and a variable amount of ligand, chosen such that the total concentration of metal, C_M , is much greater than the total concentration of ligand, C_L .
- Under these conditions we may assume that essentially all the ligand is complexed.
- The concentration of a metal–ligand complex of the general form M_xL_y is

iii. slope-ratio method ...

- Consider the reaction in which the complex M_xL_y is formed by the reaction of x moles of the cation M with y moles of a ligand L :



Mass-balance expressions for this system are

$$\blacktriangleright c_M = [M] + x[M_xL_y]$$

$$\blacktriangleright c_L = [L] + y[M_xL_y]$$

where c_M and c_L are the molar analytical concentrations of the two reactants. We now assume that, at very high analytical concentrations of L , the equilibrium is shifted far to the right and $[M] \ll x[M_xL_y]$.

iii. slope-ratio method ...

- Under this condition, the first mass-balance expression simplifies

$$\triangleright cM = x[M_xL_y]$$

- If the system obeys Beer's law, $A_1 = \epsilon b[M_xL_y] = \epsilon b c_M / x$
- where ϵ is the molar absorptivity of M_xL_y and b is the path length.
- A plot of absorbance as a function of cM is linear when there is sufficient L present to justify the assumption that $[M] \ll x[M_xL_y]$.
- The slope of this plot is $\epsilon b / x$. When c_M is made very large,
- we assume that $[L] \ll y[M_xL_y]$, and the second mass balance equation reduces

$$\triangleright cL = y[M_xL_y] \text{ and } A_2 = \epsilon b[M_xL_y] = \epsilon b c_L / y$$

iii. slope-ratio method ...

- Again, if our assumptions are valid, we find that a plot of A versus cL is linear at high concentrations of M. The slope of this line is eb/y. The ratio of the slopes of the two straight lines gives the combining ratio between M and L

$$\triangleright \frac{eb/x}{eb/y} = \frac{y}{x}$$

Spectrophotometric titrations

- In photometric method of equivalent point detection of titrations, the appearance of an absorbing species will give a linear or conc dependent change in absorbance,
- it will yield two straight lines that intersect at the equivalent point.
- There are at least three components which may absorb light: the original substance, the titrant & the resulting product (s).
- ✓ The usual procedure is to select some wave length at w/c only one component absorb.
- In spectrophotometric titration, the absorbance of the so/n at a specified λ is measured after each addition of the titrant.
- The results are plotted (A vs ml of titrant), and the end point is determined graphically.
- The point at which the two straight lines intersects the end point.

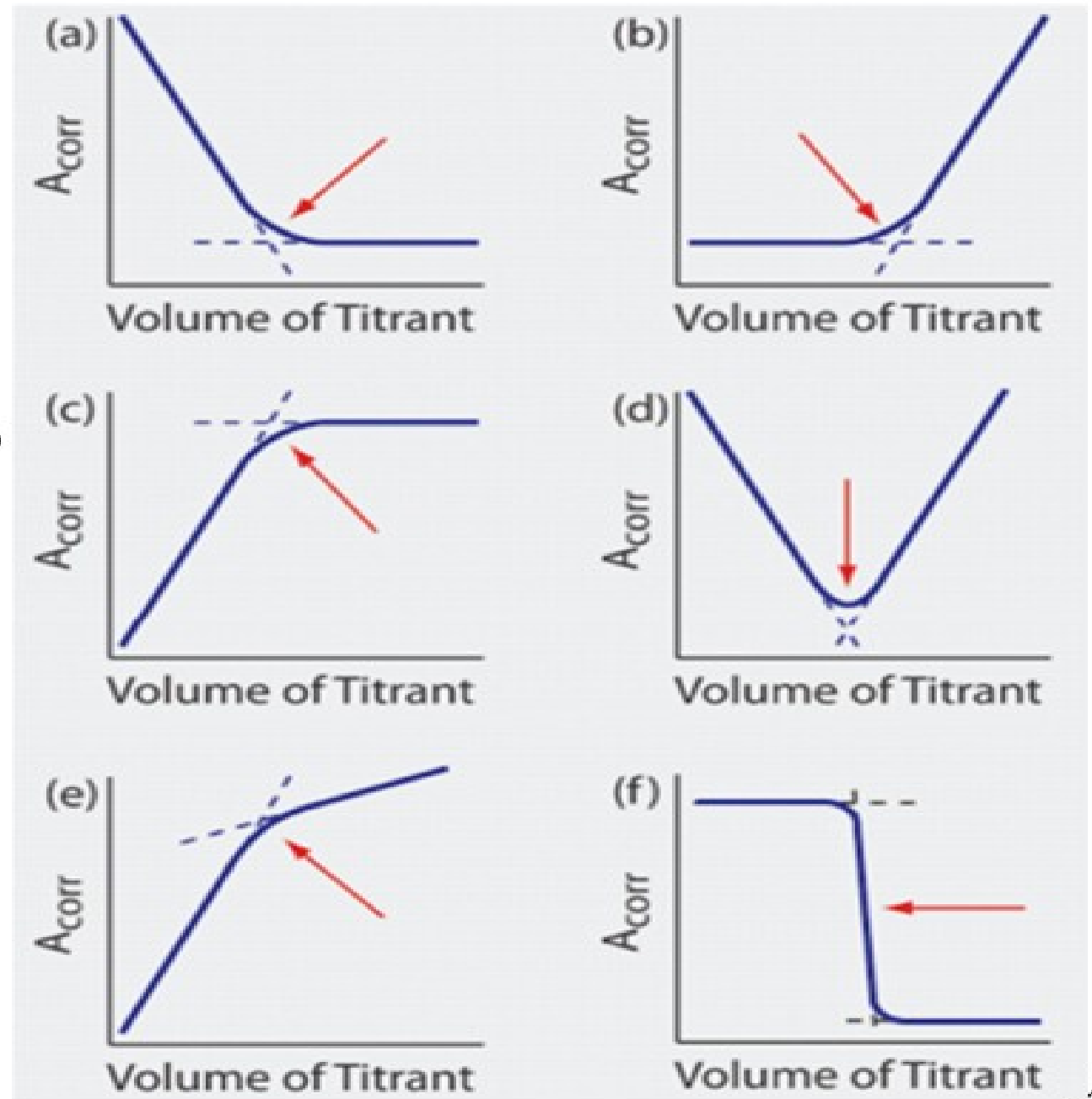
Spectrophotometric titrations...

- The graph is constructed on the basis of data obtained well before and well after the end pt.
- Some typical titration curves for the reaction $X + Y \rightarrow Z$ where X the component to be determined, Y titrant & Z the product (s) of the reaction.
- If X absorbs radiation energy at a specified λ and it is contaminated with other absorbing sub, a spectrophotometric titration can be carried out if a titrant can be found w/c react selectively with X.
- If X & the contaminant are the only species that absorb, the absorbance of the so/n will decrease as titrant is added.
- If Y is the only species in solution which absorbs, the so/n will not absorb until the end pt is reached.
- If Z is the only species in solution which absorbs, the absorbance will increase as product is formed.

➤ Spectrophotometric titration curves

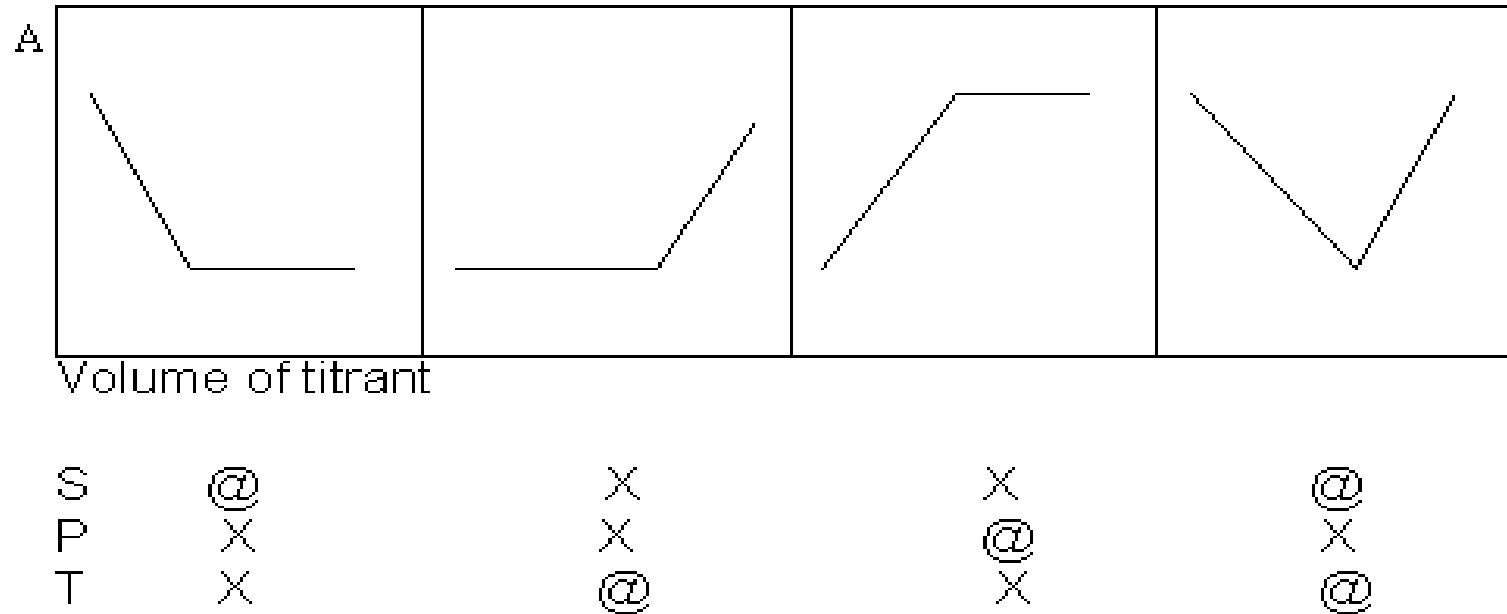
- a. Only the titrand absorbs
- b. Only the titrant absorbs
- c. Only the product absorbs
- d. Both the titrant & titrand absorb
- e. Both the product & titrant absorb
- f. Only the indicator absorbs

▪ Red arrows indicate E.Ps



Spectrophotometric titrations...

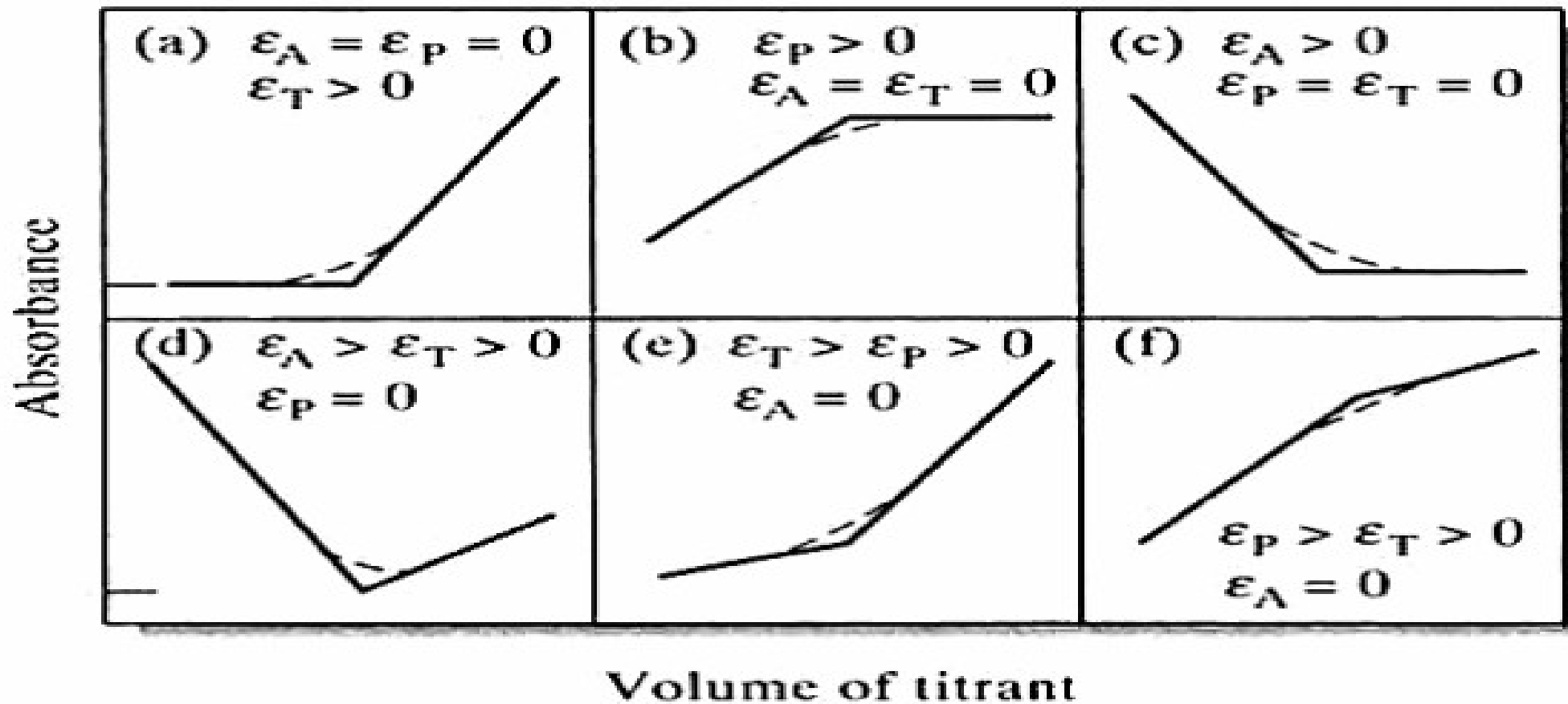
➤ The following are some of the spectrophotometric titration curves which can be observed in the normal conditions:



Where:

S = Sample , P = titration product , T = Titrant ,

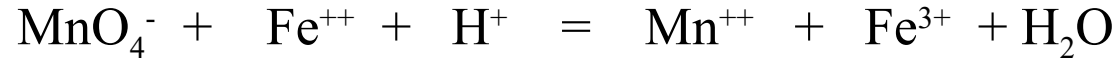
@ = Absorbing species , X = Non absorbing species.



Spectrophotometric titrations...

Examples:

1- Titration of potassium permanganate against ferrous sulfate

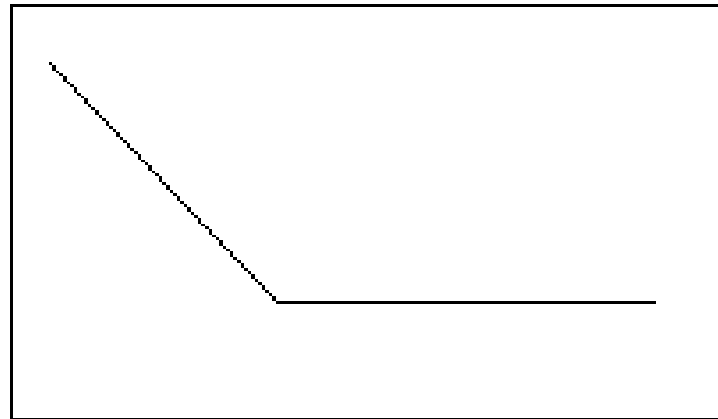


@ at 540

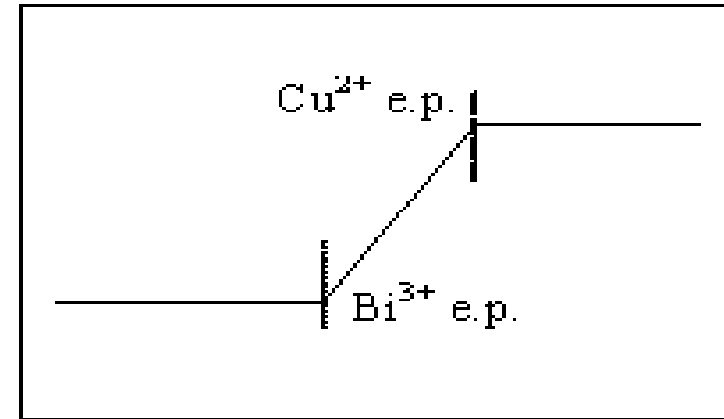
(X) at 540

(X) at 540

(X) at 540 nm



Titration of KMnO_4 with Fe^{2+} at 540 nm



Titration of a Bi^{3+} and Cu^{2+} mixture with EDTA at 745 nm

2- Titration of Bi^{3+} & Cu^{2+} mixture with EDTA;

➤ Bi^{3+} , Cu^{2+} & Bi-EDTA complex are non absorbing at 745 nm

Cu-EDTA complex is absorbing at 745 nm

Spectrophotometric titration...

Advantages

1. More accurate results than direct photometric analysis are obtained.
2. No interference from other absorbing substances because the end point depends on the change in the absorbance curve and not on the absorbance value (affect only the curve shape and sharpness of end point).
3. Can be used for titration of very dilute solutions (Sensitive).
4. Not need favorable equilibrium constants as those required for titration that depends upon observations near the end point.
5. Can be used for (applied to) all types of reactions (redox, acid-base, complexometry, pptmetryetc.)

Colorimetry

- Is a technique which involves measurement of absorbance in the visible region is known as colorimetry.
- It involves measurement of color intensity of compounds.

☐ Requirements for colorimetry

- ✓ the substance should be colored
- ✓ or it should be able to be derivatized in to colored product.
 - While derivatizing
- The reagent should be specific
- The color produced should be stable enough until the analysis is completed
- Color intensity should be directly proportional to the concentration of the analyte.

Application- colored drugs and those drugs which can be derivatized

Measuring a spectrum

- The degree of interaction of the sample with radiation (T or A) is determined by measuring both the intensity of the incident radiation with and without the sample
- Because most samples measured are in solution, the blank should be measured on a cuvette containing the pure solvent.
- This process eliminates from the sample measurement any absorbance due to the solvent.
- With a single-beam instrument, the cuvette containing the solvent is placed in the spectrophotometer, and the blank is measured.
- The sample solution is then measured in the same cuvette.

Measuring

- All modern instruments automatically store the reference *I_0 values*, which are used to *calculate* absorbance values for the sample.
- With a dual- or split-beam instrument, two cuvettes are required.
- Both cuvettes are initially filled with pure solvent, and a so-called balance measurement is performed.
- This measurement reflects the difference in absorbance between the two optical paths in use.
- The sample cuvette is then filled with the sample solution for measurement, and *I_0 and I* are measured virtually simultaneously.
- The resulting spectrum is corrected by subtracting the balance spectrum.

Thank you

...



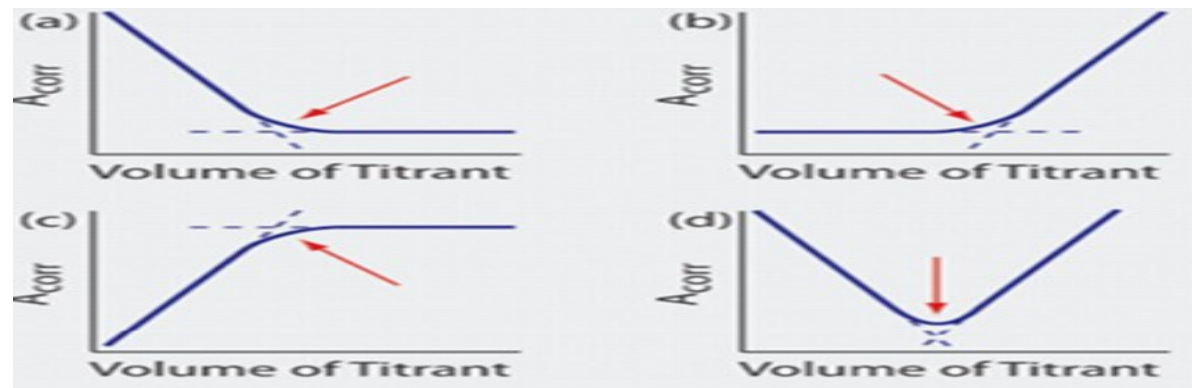
Quiz

Q1 why we apply derivatization spectra in spectrophotometric analysis ? Discuss

Q2. what are the advantages of spectrophotometric titration in analysis ?

Q3. write the application of spectrophotometry in pharmaceutical analysis

Q4. disuse according to spectrophotometric titration graphs a, b, c and d given below?



Q5. Calculate the concentration of methyl testosterone in an ethanolic solution of which the absorbance in a 1 cm cell at its λ_{max} , 241 nm, was found to be 0.890. The specific absorptivity in the B.P. is 540 at 241 nm?